CHAPTER 09

Carboxylic Acid Derivatives

Content and rendering review packet

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

Source snapshot SHA-256: faa728a8c2e404ae2641a80febcd969255ed76ca489f692ae4b40b4215fd32ed

Review all claims, molecule graphs, lessons, exercises, hints and displayed states. Keys are author claims, not ground truth. Screenshots show static states with reduced motion: they do not prove animation timing or persistence behavior.

For every finding: item ID/page; severity (critical/major/minor); exact claim; derivation or reference evidence; proposed correction; confidence. List unreviewed/uncertain items. Model agreement is not scientific verification.

Coverage checklist

ch09:exercise:cyclic-derivative | cyclic-derivative:stage-1, cyclic-derivative:stage-2, cyclic-derivative:stage-3

ch09:exercise:acyl-order | acyl-order:stage-1, acyl-order:stage-2, acyl-order:stage-3

ch09:exercise:hydrolysis-products | hydrolysis-products:stage-1, hydrolysis-products:stage-2, hydrolysis-products:stage-3

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ch09:molecule:ester | ester

ch09:molecule:betalactam | betalactam

ch09:molecule:lactone | lactone

ch09:molecule:methylacetate | methylacetate

ch09:molecule:ester-intermediate | ester-intermediate

ch09:molecule:anhyd | anhyd

ch09:molecule:acetamide | acetamide

ch09:curriculum | scope-1, scope-2, scope-3

Curriculum scope

scope-1: 9.1 Amides, Esters, and Anhydrides

scope-2: 9.2 Reactivity Principles

scope-3: 9.3 Nucleophilic Acyl Substitution Reactions

Objective: Identify esters, amides, anhydrides, lactones, and lactams

Objective: Compare acyl substitution reactivity

Objective: Predict hydrolysis and transesterification outcomes

Supplied Kaplan curriculum: Chapter 9; section scope checked against local body and summary | organic_chemistry_kaplan/OEBPS/chapter09b.xhtml

Nucleophilic acyl substitution: OpenStax 21.2, inspected | https://openstax.org/books/organic-chemistry/pages/21-2-nucleophilic-acyl-substitution-reactions

Carboxylic acid reactions and esterification: OpenStax 21.3, inspected | https://openstax.org/books/organic-chemistry/pages/21-3-reactions-of-carboxylic-acids

Acid derivative naming: OpenStax 21.1, inspected | https://openstax.org/books/organic-chemistry/pages/21-1-naming-carboxylic-acid-derivatives

No source passage is included. Request relevant authoritative excerpts when a claim cannot be checked by derivation.

families | Recognize acid derivatives

Read the atom directly attached to acyl carbon

families:rule | Core Rules

Esters have C(=O)–OR; amides have C(=O)–NR2; anhydrides have C(=O)–O–C(=O). Lactones are cyclic esters and lactams are cyclic amides. A beta-lactam contains a four-membered ring.

Visual card: Carboxylic Acid Derivative Reactivity Ski-Slope, Resonance Stabilization & IR Frequencies

Chemistry diagram: Cyclic amide versus cyclic ester. Count heteroatoms when counting ring members.

Reactivity: Acid Chloride > Anhydride > Thioester > Ester > Amide > $RCOO^-$

IR C=O: Cl (1800) > Anhyd (1810/1760) > Ester (1735) > Amide (1650)

Acid Chloride (RCOCl) | LG = Cl^- (pKaH = −7); Inductive pull > resonance | Fastest reactivity; highest IR C=O (~1800 cm^{-1})

Anhydride ($(RCO)_2O$) | LG = $RCOO^-$ (pKaH = 4.8); Shared resonance | High reactivity; TWO IR C=O peaks (~1810 & 1760 cm^{-1})

Ester (RCOOR′) | LG = RO^- (pKaH = 16); Moderate O resonance | Moderate reactivity; IR C=O ~1735 cm^{-1}

Amide ($RCONR'_2$) | LG = R_2N^- (pKaH = 38); Strongest N→C=O resonance | Least reactive neutral derivative; IR C=O ~1650 cm^{-1}

Typical acyl-derivative reactivity ladder — — Acid Chloride (RCOCl) > Acid Anhydride > Thioester (RCOSR′) > Ester (RCOOR′) > Amide ($RCONR'_2$) > Carboxylate Ion ($RCOO^-$).

Two Governing Factors: Leaving Group Basicity + Ground-State Resonance — — Cl^- (pKaH −7) is the weakest base and 3p Cl has poor π overlap with 2p C; $–NR_2$ (pKaH ~38) is a terrible leaving group and 2p N donates strongly into C=O!

IR Carbonyl Stretch Mirrors Double-Bond Character — — Stronger resonance donation into C=O weakens the C=O bond (more single-bond character → lower wavenumber): Acid Chloride (1800 cm^{-1}) > Ester (1735 cm^{-1}) > Ketone (1715 cm^{-1}) > Amide (1650 cm^{-1}).

families:cue | Memory Cues

Acyl C → O–C: ester. Acyl C → N: amide. Acyl C → O → acyl C: anhydride. For esters, name the O-alkyl group then the acid-derived -oate. Amides use -amide, with N- prefixes for N substituents. Anhydrides use the acid name(s) plus anhydride; list both acids for a mixed anhydride. In lactam/lactone shorthand, beta = four ring atoms, gamma = five, delta = six; count the ring heteroatom and carbonyl carbon, excluding the exocyclic oxygen.

Visual card: Why Are Thioesters (Acetyl-CoA) So Much More Reactive Than Oxygen Esters?

Compare Thioesters (R–CO–SR′), Oxygen Esters (R–CO–OR′), and Amides ($R–CO–NR'_2$):

Thioester vs. Oxygen Ester Orbital Overlap — 3p Sulfur Has Poor π Overlap with 2p Carbon! — Because the C–S bond is long and 3p/2p orbitals mismatch in size, sulfur gives very little resonance stabilization to C=O (leaving the carbonyl carbon highly δ+).

Thiolate (RS^-) vs. Alkoxide (RO^-) Leaving Group — RS^- (pKaH ~10) Is a Much Better Leaving Group Than RO^- (pKaH ~16) — Thioesters sit right below anhydrides in reactivity—stable enough to survive in water, yet reactive enough for enzymatic acyl transfer!

Why Does an Anhydride Show TWO C=O IR Peaks? — Symmetric (~1810 cm^{-1}) and Asymmetric (~1760 cm^{-1}) Coupled Stretches — Two C=O bonds joined through a central oxygen vibrate in-phase and out-of-phase, giving a diagnostic doublet in IR!

families:trap | Common Mistakes / MCAT Traps

An ether has no adjacent carbonyl. An N-substituted amide is still an amide even if it has no N–H bond. Ring size names count heteroatoms as ring members. Ammonia, primary and secondary amines can give ordinary neutral amides by acylation and N–H loss. Tertiary amines lack that N–H: they can act as bases or form acylammonium intermediates, rather than the analogous neutral amide by simple acylation.

Visual card: Trying to Convert an Amide or Ester Directly into an Acid Chloride or Anhydride

Common Trap — — Reacting acetamide with NaCl expecting acetyl chloride, or reacting ethyl acetate with acetate ion expecting acetic anhydride.

Correct Reasoning — — The reactivity ladder predicts many unactivated substitutions. Activation, coupling reagents, catalysis and equilibrium control can enable other conversions; downhill is not an absolute law.

families:app | MCAT Applications

Triacylglycerols contain ester links; proteins contain amide links. Beta-lactam antibiotics exploit a strained amide structure.

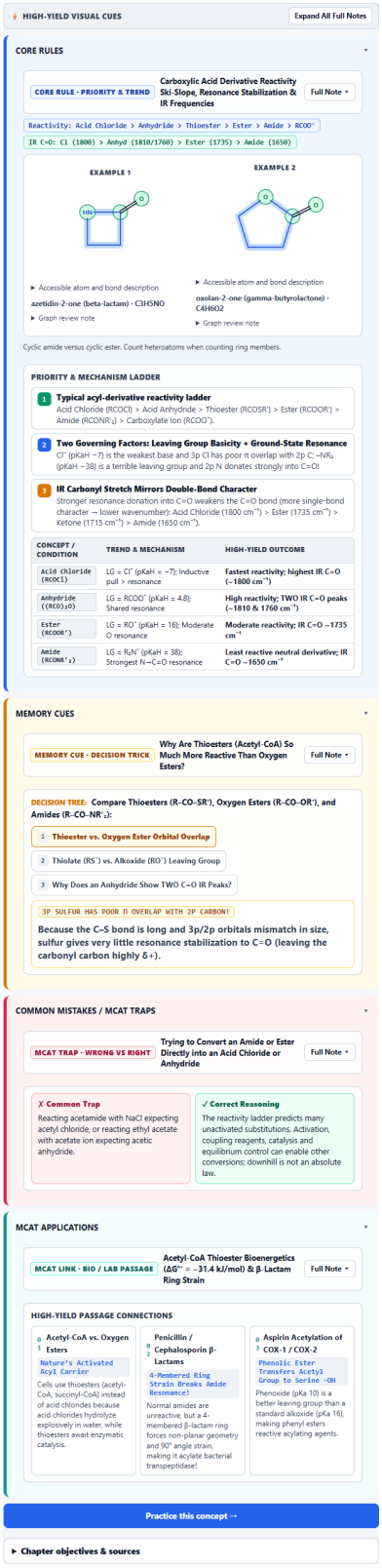
Visual card: Acetyl-CoA Thioester Bioenergetics ($\Delta G^{\circ\prime} = -31.4$ kJ/mol) & β-Lactam Ring Strain

Acetyl-CoA vs. Oxygen Esters — Nature's Activated Acyl Carrier — Cells use thioesters (acetyl-CoA, succinyl-CoA) instead of acid chlorides because acid chlorides hydrolyze explosively in water, while thioesters await enzymatic catalysis.

Penicillin / Cephalosporin β-Lactams — 4-Membered Ring Strain Breaks Amide Resonance! — Normal amides are unreactive, but a 4-membered β-lactam ring forces non-planar geometry and 90° angle strain, making it acylate bacterial transpeptidase!

Aspirin Acetylation of COX-1 / COX-2 — Phenolic Ester Transfers Acetyl Group to Serine –OH — Phenoxide (pKa 10) is a better leaving group than a standard alkoxide (pKa 16), making phenyl esters reactive acylating agents.

Screenshot: families-visuals



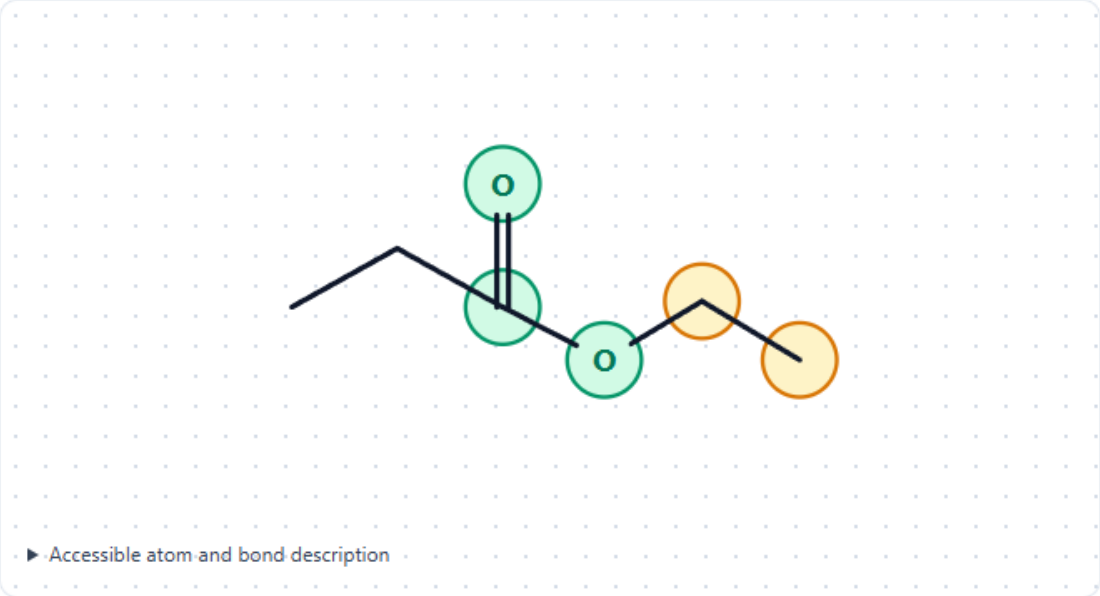
families:step-1 | 1. Acyclic ester

Ethyl propanoate connects the acyl carbon to oxygen. The alkoxy carbon chain lies beyond that oxygen. An amide instead bonds the acyl carbon directly to nitrogen.

Displayed molecule ID: ester

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Screenshot: families-step-1



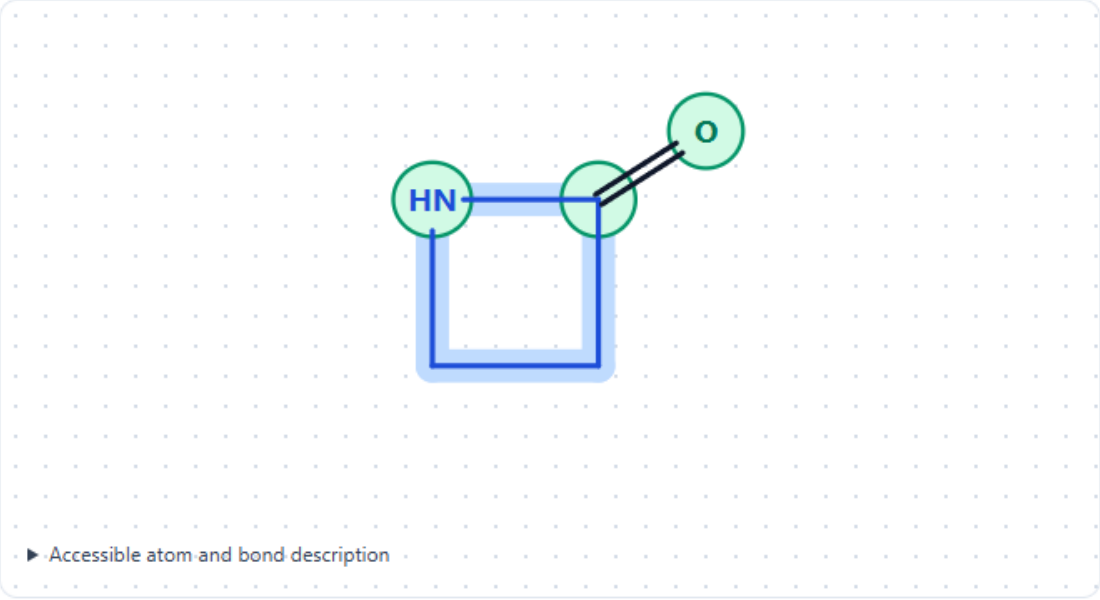
families:step-2 | 2. Cyclic amide

The beta-lactam has four ring atoms, one of them N. The carbonyl carbon is part of the ring and the oxygen is outside it.

Displayed molecule ID: betalactam

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Screenshot: families-step-2



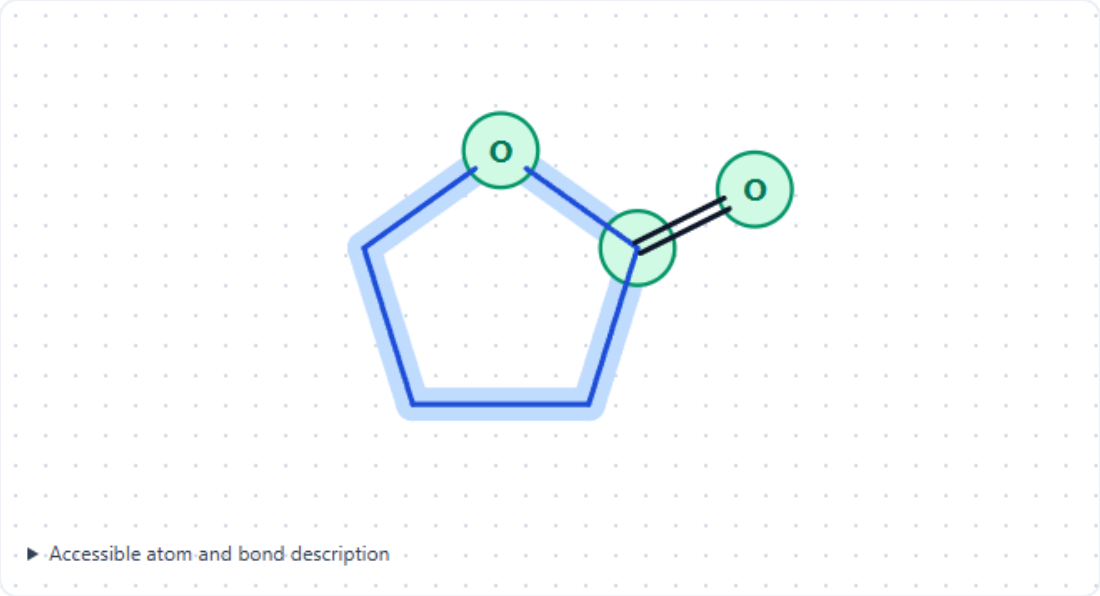
families:step-3 | 3. Cyclic ester

A lactone contains an ester linkage within a ring. This five-membered lactone has one ring oxygen and four ring carbons.

Displayed molecule ID: lactone

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Screenshot: families-step-3



acyl-mechanism | Acyl substitution and relative reactivity

Addition → tetrahedral intermediate → elimination

acyl-mechanism:rule | Core Rules

A nucleophile adds to an acyl carbon and shifts the pi pair to O. Collapse of the tetrahedral intermediate reforms C=O and expels a leaving group. For comparable unactivated derivatives, a useful order is acid chloride > anhydride > ester > amide. Leaving-group basicity and resonance donation both matter. Inductive withdrawal transmits through sigma bonds and can increase acyl-carbon electrophilicity. Steric crowding can hinder nucleophile approach. These effects compete with leaving-group ability, resonance and catalysis; a single naming-priority or conjugation shortcut cannot rank every reaction.

Visual card: Nucleophilic Acyl Substitution Mechanism & Organometallic / Hydride Selectivity

Chemistry diagram: Hydroxide addition gives a tetrahedral intermediate; subsequent collapse is a separate step. Schematic, not balanced.

Step 1: Addition (sp^2 → sp^3)

Step 2: Elimination (Restore C=O)

Best LG Leaves Tetrahedral Int

RMgBr (2 Eq) + Ester → 3° Alcohol

Ester + 2 Eq RMgBr | Cannot stop at ketone! | Gives 3° Alcohol with TWO identical R groups from Grignard

Acid Chloride + 1 Eq R_2CuLi | Gilman reagent stops at KETONE! | Soft cuprate reacts with RCOCl but NOT with the ketone product

Ester + DIBAL-H (−78 °C) | 1 hydride equivalent stops at ALDEHYDE! | Tetrahedral aluminate intermediate stays frozen until H_3O^+ workup

Two-Step Addition–Elimination Pathway — — Step 1: Nucleophile attacks sp^2 C=O to form a tetrahedral sp^3 intermediate. Step 2: Carbonyl π bond reforms and kicks out the WEAKEST base (best leaving group).

Grignard / Organolithium + Ester or Acid Chloride (2 Equivalents!) — — First equivalent forms a ketone in situ; because a ketone is MORE reactive than an ester, a second RMgBr immediately attacks to yield a tertiary alcohol ($R'–C(OH)R_2$).

Stopping at the Ketone or Aldehyde — — Acid Chloride + R_2CuLi (Gilman) stops at the KETONE; Ester + DIBAL-H at −78 °C (or Acid Chloride + $LiAl(Ot\text{-}Bu)_3H$) stops at the ALDEHYDE.

acyl-mechanism:cue | Memory Cues

Addition–elimination replaces the acyl substituent while retaining a carbonyl.

Visual card: Predicting the Product of Hydride or Organometallic Addition to Acyl Derivatives

Which derivative and nucleophile are reacting?

Ester (RCOOR′) + Excess RMgBr or $LiAlH_4$ — Cleaves –OR′! Gives 3° Alcohol (with 2 RMgBr) or 1° Alcohol + R′OH (with $LiAlH_4$) — Do NOT forget that the leaving group alcohol (R′OH) is also released as a second organic product!

Amide ($RCONR'_2$) + $LiAlH_4$ (then H_2O) — C=O Oxygen Leaves Instead of NR'_2! → Yields AMINE ($RCH_2NR'_2$) — Because R_2N^- is a terrible leaving group, aluminum coordinates to oxygen and expels O^{2-}/Al-oxide via an iminium ion, deoxygenating C=O to CH_2!

Nitrile (R–C≡N) + 1 Eq RMgBr (then H_3O^+) vs. $LiAlH_4$ — RMgBr → Ketone (R–CO–R′ via Imine Hydrolysis); $LiAlH_4$ → 1° Amine (RCH_2NH_2) — Grignard attacks C≡N once to form a negatively charged imine salt (R–C(=NMgBr)R′) that cannot be attacked a second time and hydrolyzes to a ketone!

acyl-mechanism:trap | Common Mistakes / MCAT Traps

Aldehydes and ketones normally add rather than substitute because hydride or an ordinary carbanion would be a poor leaving group. Conjugation means adjacent aligned p orbitals delocalize electrons; it can stabilize a reactant, intermediate or transition state. A rate depends on the activation free-energy difference, so stabilization alone does not prove a faster reaction. Enones also offer beta attack (1,4-addition), which differs from direct carbonyl attack (1,2-addition). Heteroatom lone-pair donation into C=O, especially from amide N, reduces carbonyl electrophilicity. The effect of conjugation requires the actual substrate, nucleophile, site and conditions.

Visual card: Stopping Ester + Grignard at the Ketone or Cleaving the C–N Bond in Amide + $LiAlH_4$

Common Trap — — Reacting ethyl benzoate with 1 eq CH_3MgBr to get acetophenone cleanly, or reducing an amide with $LiAlH_4$ to get an alcohol + amine.

Correct Reasoning — — Ketones are more reactive than esters, so Ester + RMgBr always double-adds to a 3° alcohol; Amide + $LiAlH_4$ keeps the C–N bond intact and reduces C=O to CH_2 (forming an amine)!

acyl-mechanism:app | MCAT Applications

Amide resonance makes peptide bonds relatively resistant to uncatalyzed hydrolysis. Strained beta-lactams are more reactive than ordinary unstrained amides. Small rings can have angle strain and torsional strain. Beta-lactam geometry can reduce optimal amide orbital overlap and hydrolysis can relieve strain; the extent of nitrogen pyramidalization and resonance loss depends on ring fusion/substitution. The displayed unfused four-membered connectivity diagram is not a 3D bond-angle model.

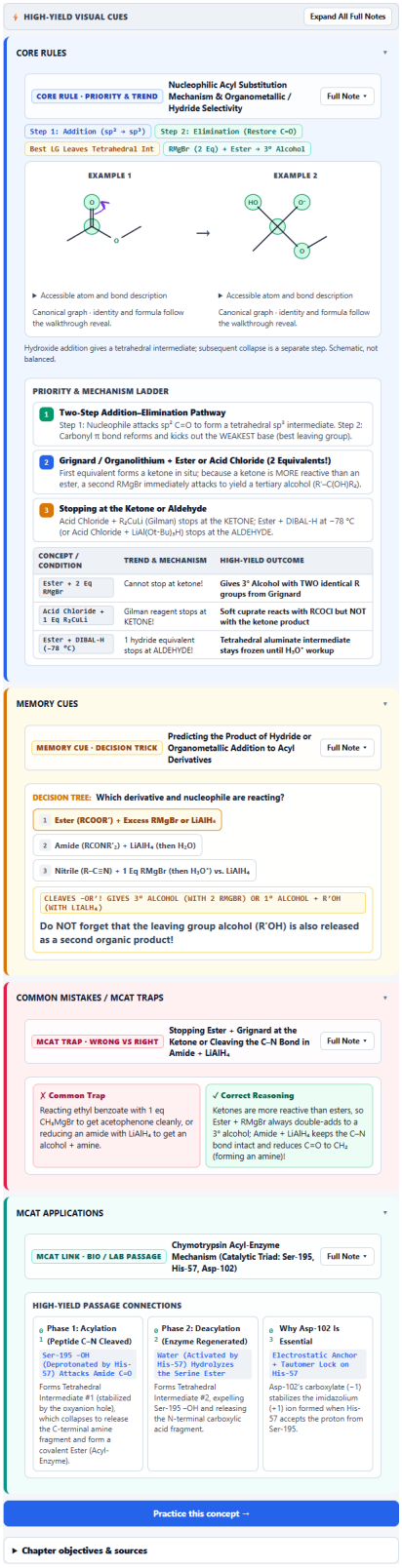
Visual card: Chymotrypsin Acyl-Enzyme Mechanism (Catalytic Triad: Ser-195, His-57, Asp-102)

Phase 1: Acylation (Peptide C–N Cleaved) — Ser-195 –OH (Deprotonated by His-57) Attacks Amide C=O — Forms Tetrahedral Intermediate #1 (stabilized by the oxyanion hole), which collapses to release the C-terminal amine fragment and form a covalent Ester (Acyl-Enzyme).

Phase 2: Deacylation (Enzyme Regenerated) — Water (Activated by His-57) Hydrolyzes the Serine Ester — Forms Tetrahedral Intermediate #2, expelling Ser-195 –OH and releasing the N-terminal carboxylic acid fragment.

Why Asp-102 Is Essential — Electrostatic Anchor + Tautomer Lock on His-57 — Asp-102's carboxylate (−1) stabilizes the imidazolium (+1) ion formed when His-57 accepts the proton from Ser-195.

Screenshot: acyl-mechanism-visuals



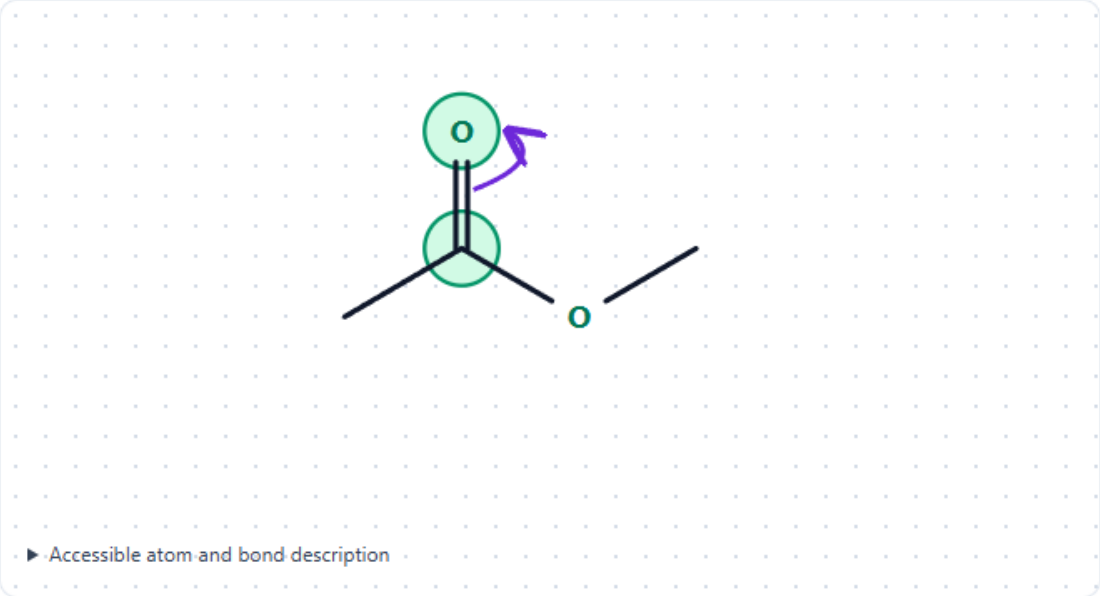
acyl-mechanism:step-1 | 1. Add to the carbonyl

Hydroxide attacks the ester carbonyl carbon while the pi pair moves to oxygen. The arrow shows that pi-pair movement, not the entire mechanism.

Displayed molecule ID: methylacetate

{"highlights": {"prin": [1, 2], "flow": [{"fromBond": 1, "toAtom": 2}]}, "revealName": false}

Screenshot: acyl-mechanism-step-1



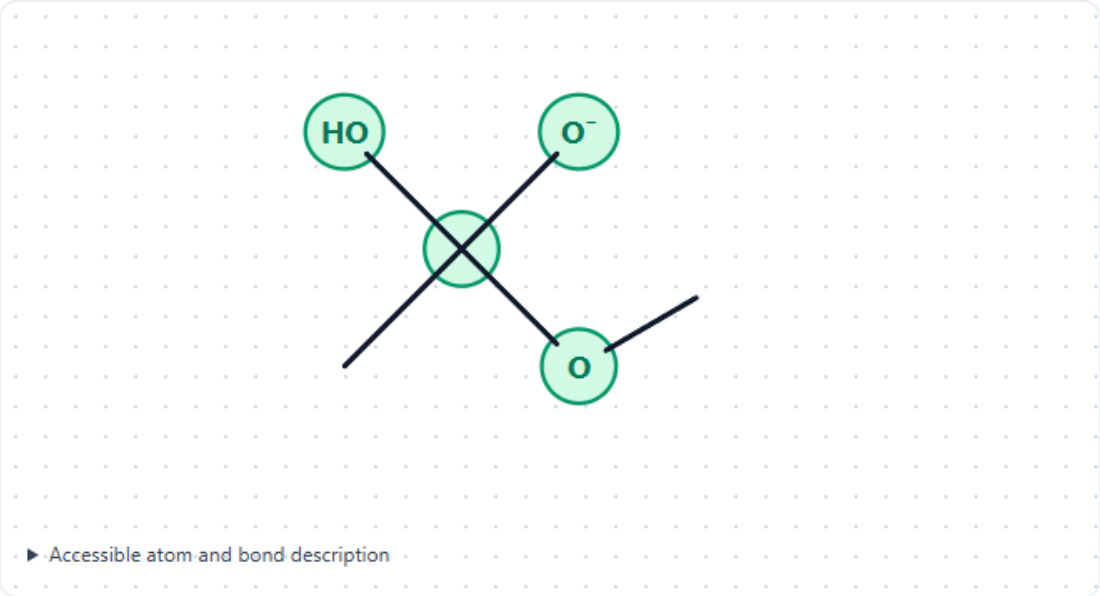
acyl-mechanism:step-2 | 2. Tetrahedral intermediate

The carbon now bears the incoming OH, the original OCH3, and an O− derived from the carbonyl oxygen. It has four single bonds. Four directions are spread in a 2D projection for readability, not a claim of 90-degree physical bond angles.

Displayed molecule ID: ester-intermediate

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Screenshot: acyl-mechanism-step-2



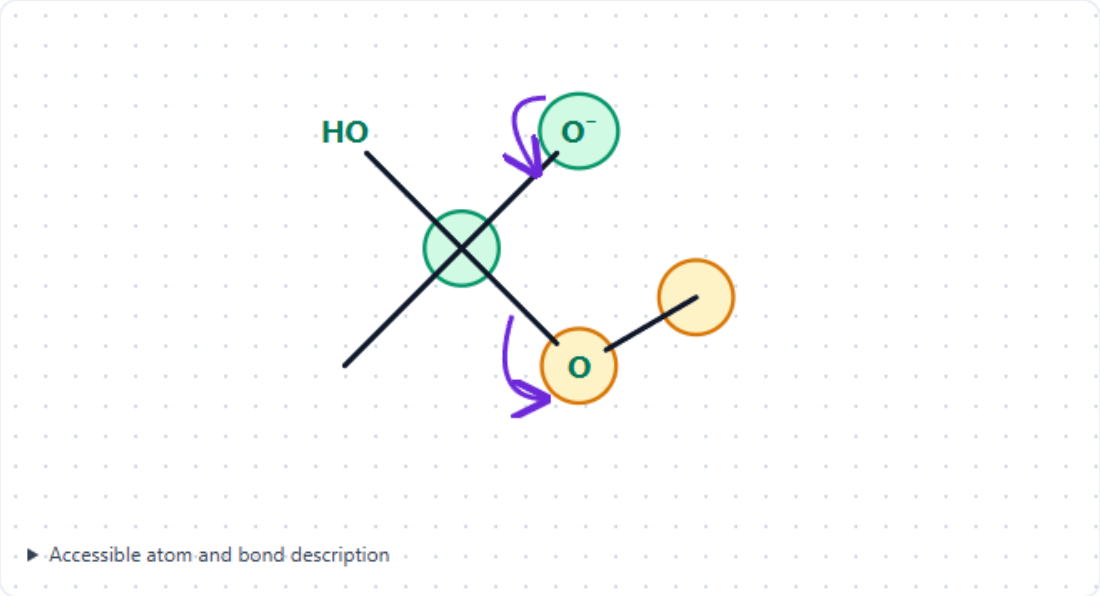
acyl-mechanism:step-3 | 3. Collapse

An oxygen lone pair reforms C=O, and the C–OCH3 bond pair moves to the leaving oxygen. Subsequent proton transfer yields carboxylate plus methanol under basic conditions.

Displayed molecule ID: ester-intermediate

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Screenshot: acyl-mechanism-step-3



▶ Accessible atom and bond description

derivative-reactions | Hydrolysis, alcoholysis, and transesterification

Conditions determine the protonation state

derivative-reactions:rule | Core Rules

Anhydrides react with water to give acids, with alcohols to give ester plus acid, and with amines to give amide plus acid-derived products. Transesterification exchanges one OR group for another under suitable catalysis. Acidic amide hydrolysis gives carboxylic acid and protonated amine; basic hydrolysis gives carboxylate and amine. A condensation joins reactants while losing a small molecule. Fischer esterification is acid + alcohol ⇌ ester + water under acid catalysis; removing water or using excess alcohol favors ester. Water need not be completely absent, and aqueous acid promotes the reverse hydrolysis.

Visual card: Hydrolysis, Alcoholysis, Transesterification, Aminolysis & Cyclic Derivatives (Lactones/Lactams)

Chemistry diagram: Separate anhydride and amide examples, not a reactant/product pair.

Hydrolysis (H_2O) → RCOOH

Alcoholysis (ROH) → Ester

Aminolysis (2 Eq RNH_2) → Amide

Lactone = Cyclic Ester | Lactam = Cyclic Amide

Acid Chloride + R′OH / Pyridine | Fastest quantitative ester synthesis | Pyridine scavenges HCl byproduct so acid-sensitive groups survive

Transesterification (RCOOR′ + R″OH) | Swap ester alkoxy group (H^+ or $R''O^-$) | Use matched base $R''O^-$ with R″OH solvent to prevent scrambling!

Anhydride + ROH or RNH_2 | 1 acyl group becomes product; 1 becomes $RCOO^-$ | Acetic anhydride cleanly acetylates salicylic acid → Aspirin

Universal Hydrolysis Endpoint — — Every carboxylic acid derivative (acid chloride, anhydride, ester, amide, nitrile) hydrolyzes in hot aqueous acid (H_3O^+, Δ) to a carboxylic acid (RCOOH) or in base (HO^-, Δ) to a carboxylate ($RCOO^-$).

Transesterification & Matched Alkoxide Rule — — RCOOR′ + R″OH (cat. H^+ or $R''O^-$) ⇌ RCOOR″ + R′OH. Whenever using a basic alkoxide catalyst or solvent with an ester, match the alkoxide ($R'O^-$/R′OH) unless you WANT transesterification!

Why Aminolysis of Acid Chlorides Needs 2 Equivalents of Amine — — RCOCl + 2 $R'NH_2$ → RCONHR′ + $R'NH_3^+Cl^-$. One equivalent forms the amide; a second equivalent of amine (or added pyridine/Et_3N/NaOH) neutralizes the HCl generated!

derivative-reactions:cue | Memory Cues

Follow the acyl fragment and the departing group separately. Saponification is basic ester cleavage; a triacylglycerol yields glycerol and fatty-acid salts. Suitable dicarboxylic acids can dehydrate on heating to five- or six-membered cyclic anhydrides when geometry and conditions allow; not every diacid cyclizes.

Visual card: Cyclic vs. Acyclic Derivative Recognition & Reaction Conditions

Which interconversion or cyclic derivative are you analyzing?

Hydroxy Acid (HO–(CH_2)_n–COOH) + Acid Catalyst — Intramolecular Fischer Esterification → Cyclic Ester (LACTONE) — 5-membered (γ-lactone) and 6-membered (δ-lactone) rings form spontaneously upon mild heating in acid.

Amino Acid (H_2N–(CH_2)_n–COOH) Ring Closure — Cyclic Amide (LACTAM) — β-lactams (4-membered), γ-lactams (5-membered), and δ-lactams (6-membered); hydrolyzing a lactam opens the ring into a single linear amino acid!

Nitrile (R–C≡N) Mild vs. Vigorous Hydrolysis — Mild H_3O^+/HO^- → Primary Amide ($RCONH_2$) → Vigorous H_3O^+/Δ → Carboxylic Acid (RCOOH + NH_4^+) — Nitriles are masked carboxylic acids: hydrolysis passes through the primary amide

intermediate before yielding RCOOH.

derivative-reactions:trap | Common Mistakes / MCAT Traps

In basic ester hydrolysis, carboxylate formation helps drive the net reaction. Do not draw a free carboxylic acid as the major product until acid workup. Amide hydrolysis usually needs more forcing conditions than ester hydrolysis. Acetal/ketal protection masks an aldehyde or ketone electronically and can tolerate many bases or hydrides; aqueous acid removes it. It does not make a carbonyl universally inert or merely block approach by steric bulk.

Visual card: Using Unmatched NaOEt/EtOH on a Methyl Ester or Forgetting Cyclic Hydrolysis Gives 1 Molecule

Common Trap — — Using $NaOCH_2CH_3$ in ethanol as the base for a Claisen reaction on a METHYL ester (getting a scrambled methyl/ethyl ester mixture), or splitting a lactone into two separate fragments upon hydrolysis.

Correct Reasoning — — Always match the base/solvent to the ester (NaOMe/MeOH for methyl esters); hydrolyzing a cyclic ester (lactone) or cyclic amide (lactam) opens the ring into ONE linear molecule containing both groups!

derivative-reactions:app | MCAT Applications

Proteases catalyze peptide hydrolysis under mild biological conditions by activating nucleophiles and stabilizing intermediates; the uncatalyzed reaction is much slower.

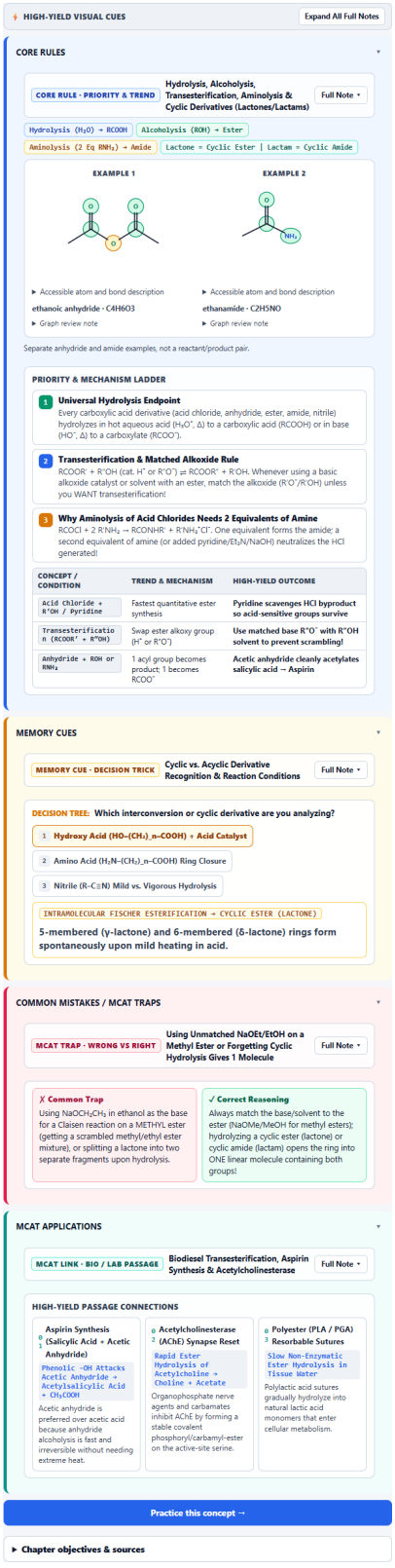
Visual card: Biodiesel Transesterification, Aspirin Synthesis & Acetylcholinesterase

Aspirin Synthesis (Salicylic Acid + Acetic Anhydride) — Phenolic –OH Attacks Acetic Anhydride → Acetylsalicylic Acid + CH_3COOH — Acetic anhydride is preferred over acetic acid because anhydride alcoholysis is fast and irreversible without needing extreme heat.

Acetylcholinesterase (AChE) Synapse Reset — Rapid Ester Hydrolysis of Acetylcholine → Choline + Acetate — Organophosphate nerve agents and carbamates inhibit AChE by forming a stable covalent phosphoryl/carbamyl-ester on the active-site serine.

Polyester (PLA / PGA) Resorbable Sutures — Slow Non-Enzymatic Ester Hydrolysis in Tissue Water — Polylactic acid sutures gradually hydrolyze into natural lactic acid monomers that enter cellular metabolism.

Screenshot: derivative-reactions-visuals



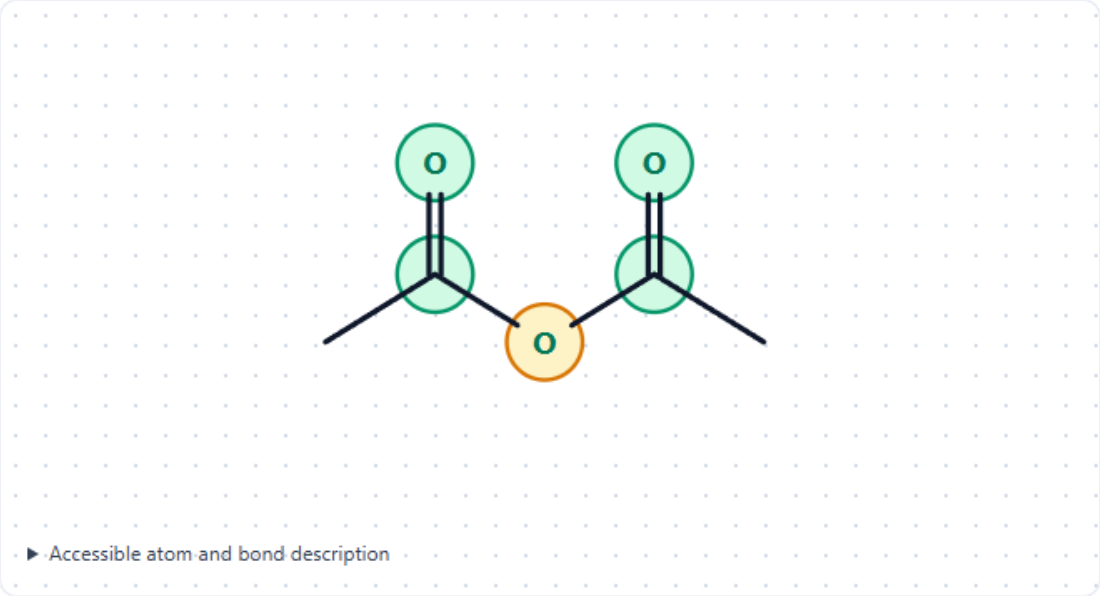
derivative-reactions:step-1 | 1. Anhydride cleavage

Water attack at either equivalent acyl carbon of ethanoic anhydride leads overall to two ethanoic acid molecules.

Displayed molecule ID: anhyd

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Screenshot: derivative-reactions-step-1



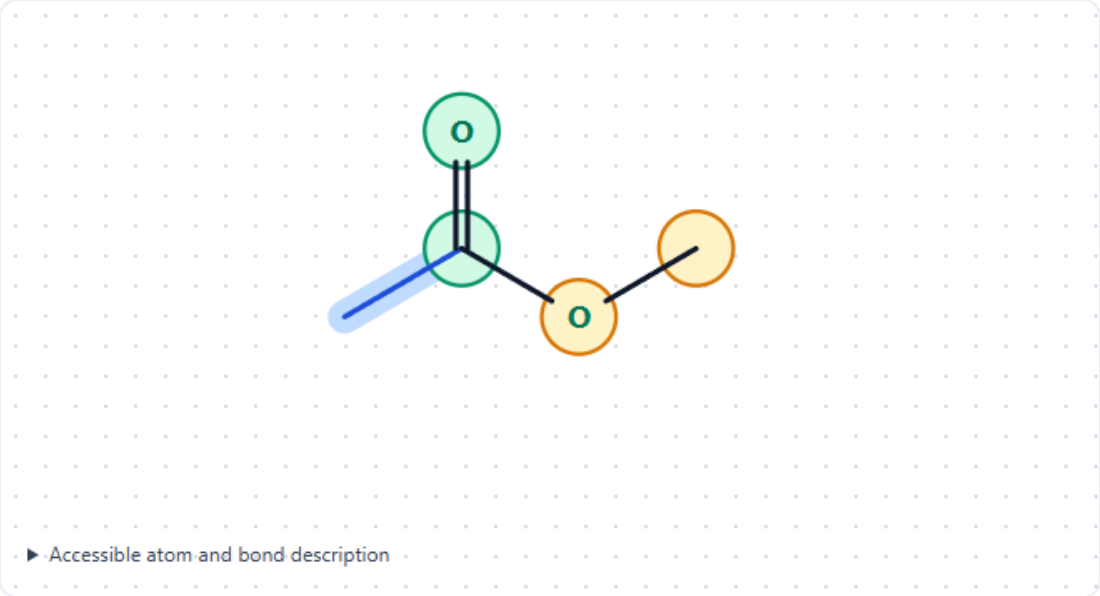
derivative-reactions:step-2 | 2. Exchange an ester group

In transesterification, the O-alkyl group can be replaced by the incoming alcohol-derived group while the acyl carbon skeleton remains.

Displayed molecule ID: methylacetate

{"highlights": {"path": [0, 1], "prin": [1, 2], "sub": [3, 4]}, "revealName": true}

Screenshot: derivative-reactions-step-2



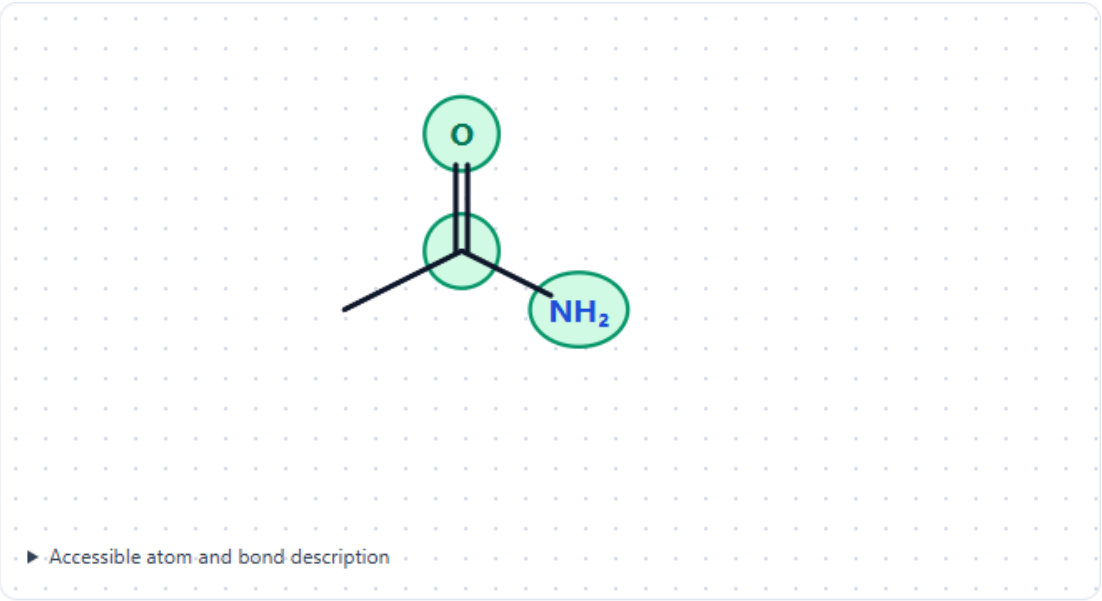
derivative-reactions:step-3 | 3. Hydrolyze an amide

An amide's N donation lowers electrophilicity and its leaving group is poor unless conditions enable departure. Strong acid/base with heating or a catalyst can accomplish hydrolysis.

Displayed molecule ID: acetamide

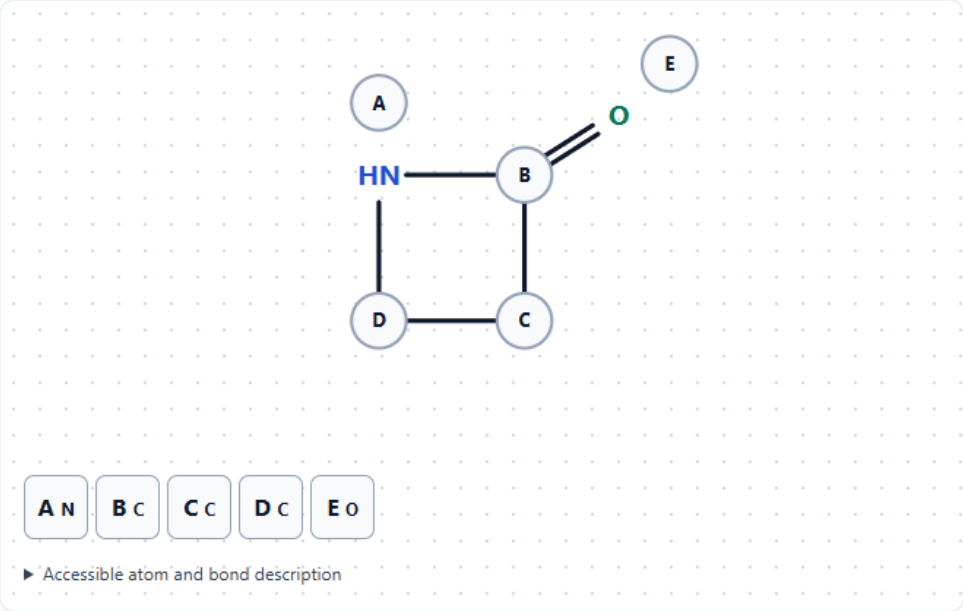
{"highlights": {"prin": [1, 2, 3]}, "revealName": true}

Screenshot: derivative-reactions-step-3



cyclic-derivative | Identify a strained derivative

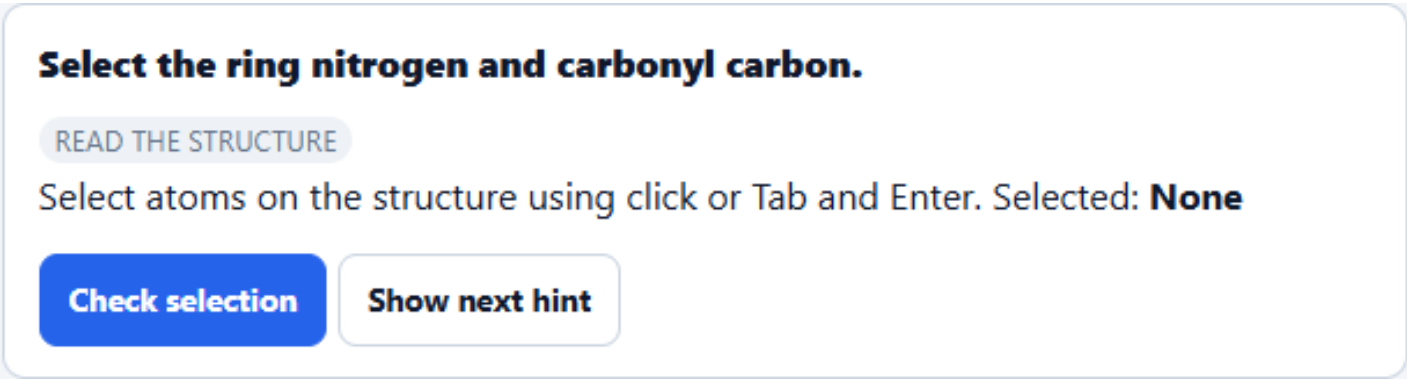
Screenshot: cyclic-derivative-stage-1-molecule



cyclic-derivative:stage-1 | Select the ring nitrogen and carbonyl carbon.

Answer type: atoms

Screenshot: cyclic-derivative-stage-1-question



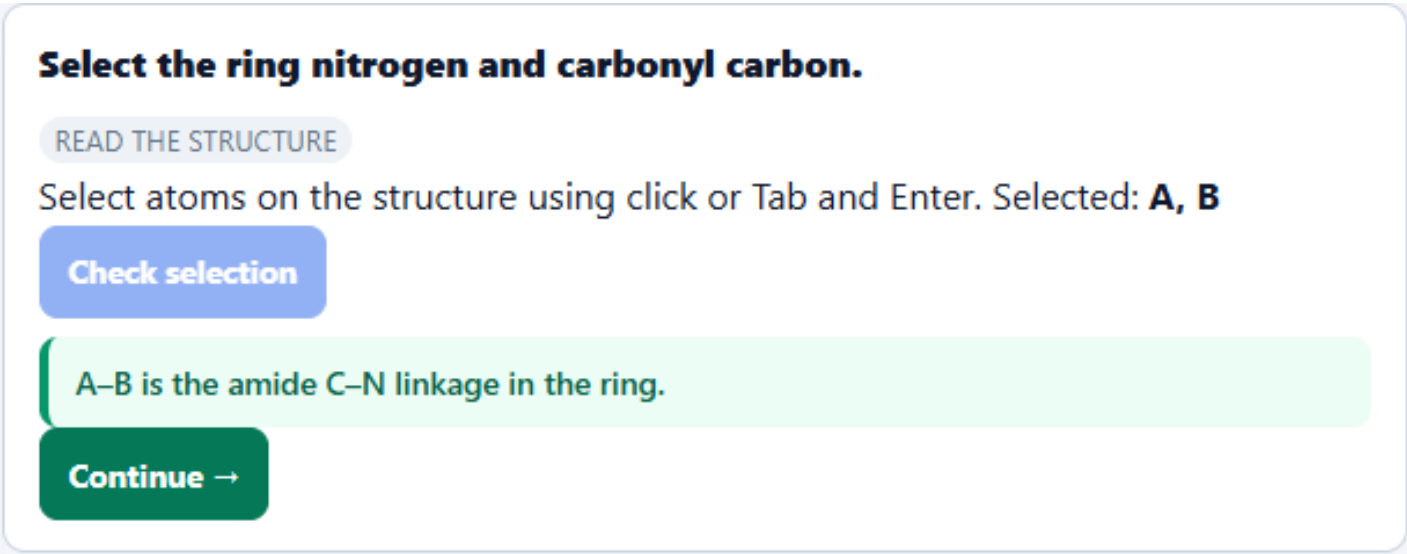
{"authorAnswer": [0, 1], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [0, 1]}}

Hint 1: Find N bonded directly to C=O carbon.

Hint 2: Select both endpoints of that bond.

Explanation: A–B is the amide C–N linkage in the ring.

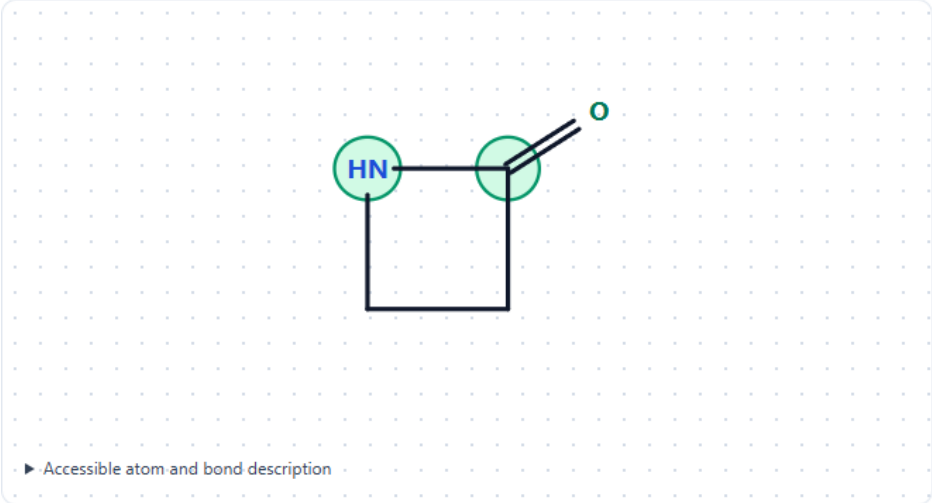
Screenshot: cyclic-derivative-stage-1-feedback



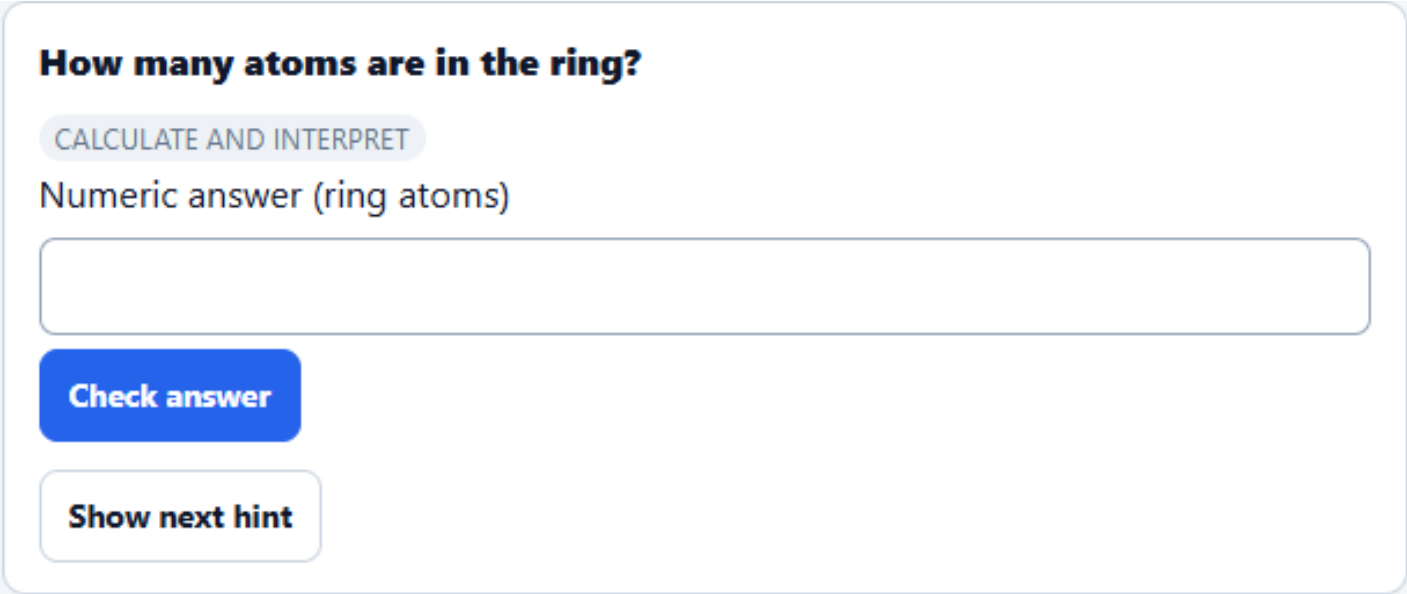
cyclic-derivative:stage-2 | How many atoms are in the ring?

Answer type: number | unit: ring atoms

Screenshot: cyclic-derivative-stage-2-molecule



Screenshot: cyclic-derivative-stage-2-question



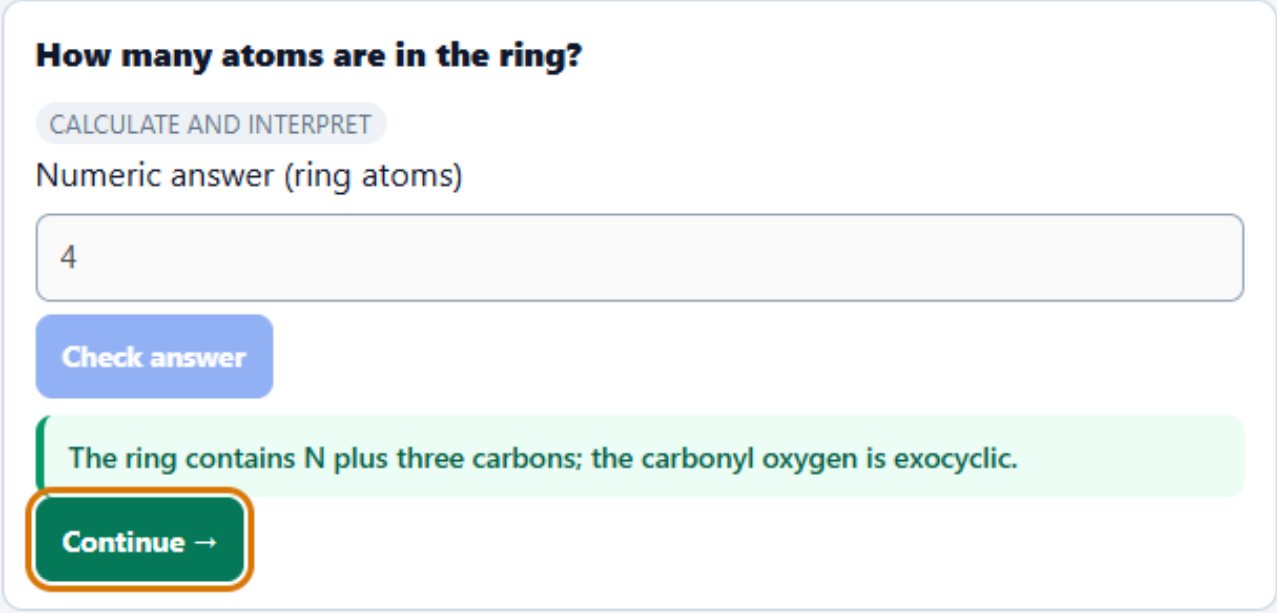
{"authorAnswer": 4, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Follow the closed path.

Hint 2: Do not count the oxygen outside the ring.

Explanation: The ring contains N plus three carbons; the carbonyl oxygen is exocyclic.

Screenshot: cyclic-derivative-stage-2-feedback



cyclic-derivative:stage-3 | The functional-group family is...

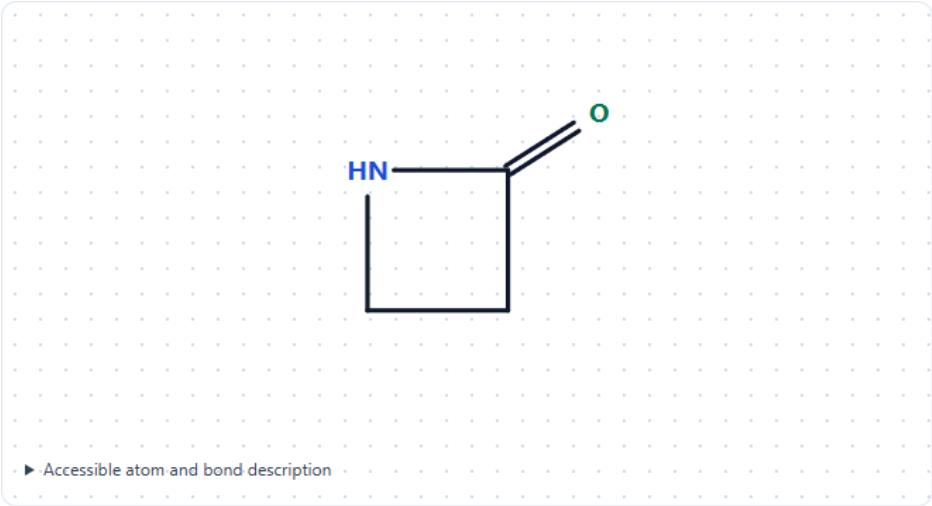
Answer type: choice

Option 0 (A): Lactam

Option 1 (B): Lactone

Option 2 (C): Ether

Screenshot: cyclic-derivative-stage-3-molecule



Screenshot: cyclic-derivative-stage-3-question



{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Use the heteroatom attached to the acyl carbon.

Hint 2: N indicates an amide.

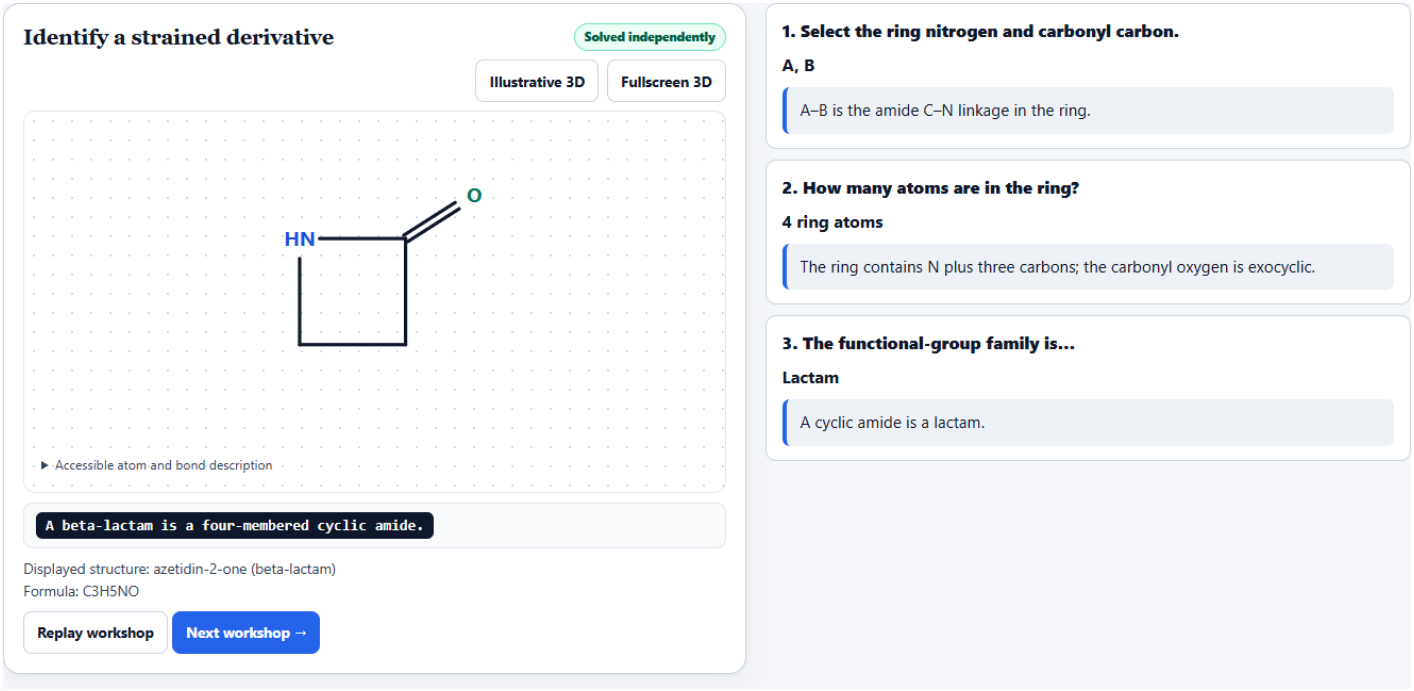
Explanation: A cyclic amide is a lactam.

Screenshot: cyclic-derivative-stage-3-feedback



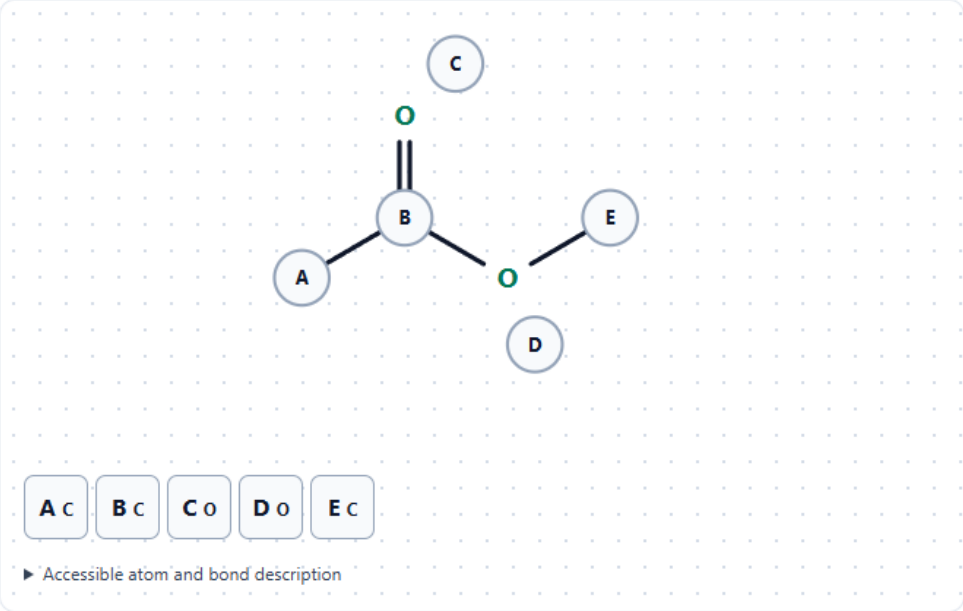
Completion summary: A beta-lactam is a four-membered cyclic amide.

Screenshot: cyclic-derivative-complete



acyl-order | Follow addition–elimination

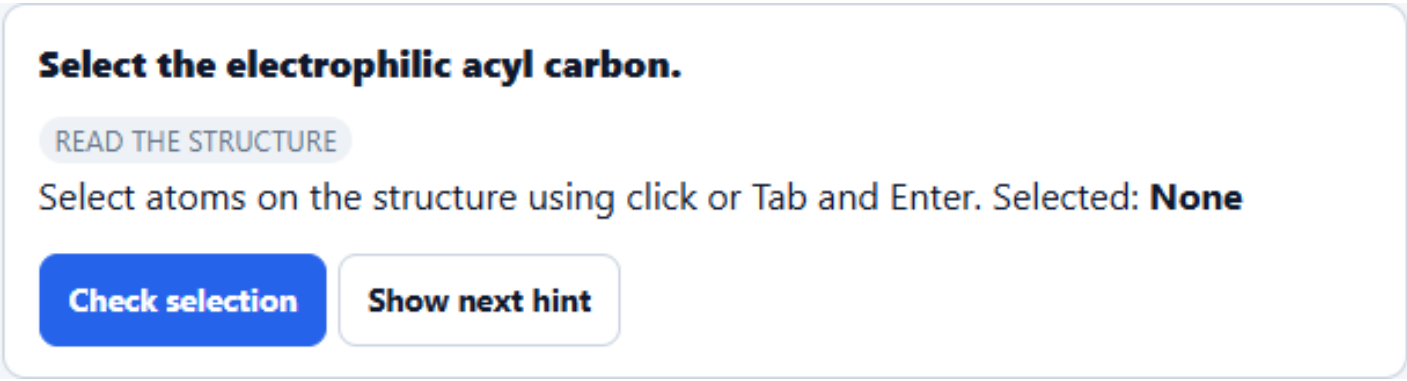
Screenshot: acyl-order-stage-1-molecule



acyl-order:stage-1 | Select the electrophilic acyl carbon.

Answer type: atoms

Screenshot: acyl-order-stage-1-question



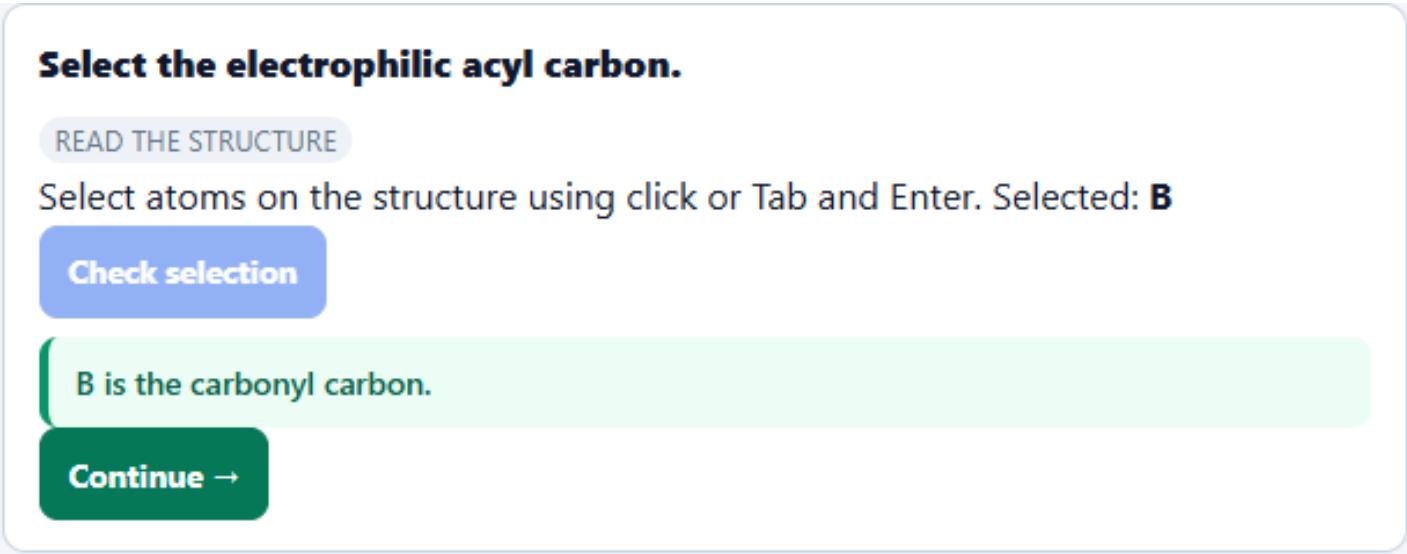
{"authorAnswer": [1], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1, 2]}}

Hint 1: Find C=O.

Hint 2: Choose carbon, not oxygen.

Explanation: B is the carbonyl carbon.

Screenshot: acyl-order-stage-1-feedback



acyl-order:stage-2 | After addition, what restores the carbonyl?

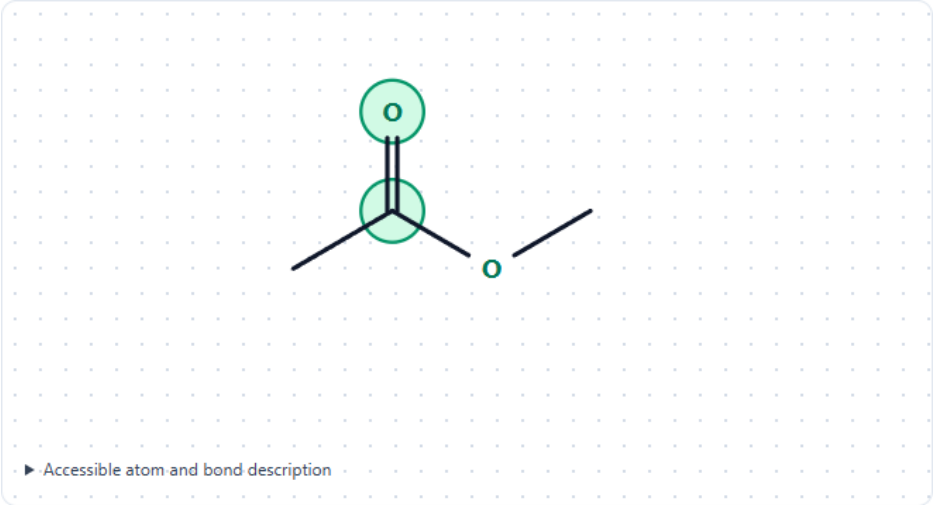
Answer type: choice

Option 0 (A): Collapse of a tetrahedral intermediate with leaving-group departure

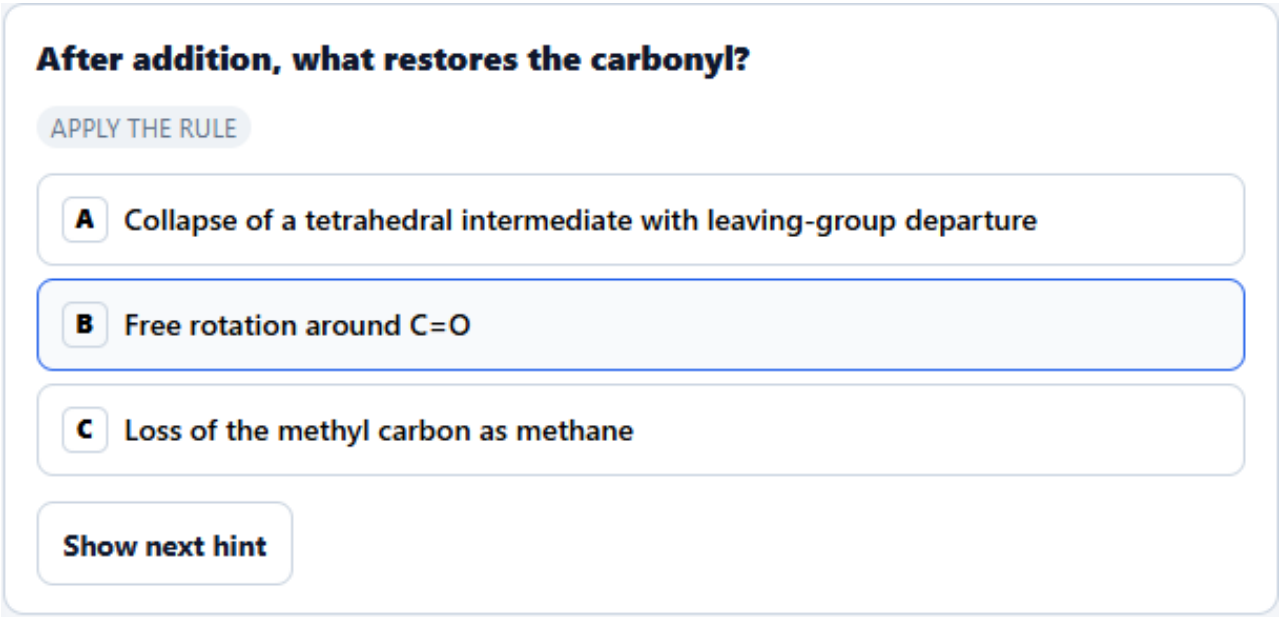
Option 1 (B): Free rotation around C=O

Option 2 (C): Loss of the methyl carbon as methane

Screenshot: acyl-order-stage-2-molecule



Screenshot: acyl-order-stage-2-question



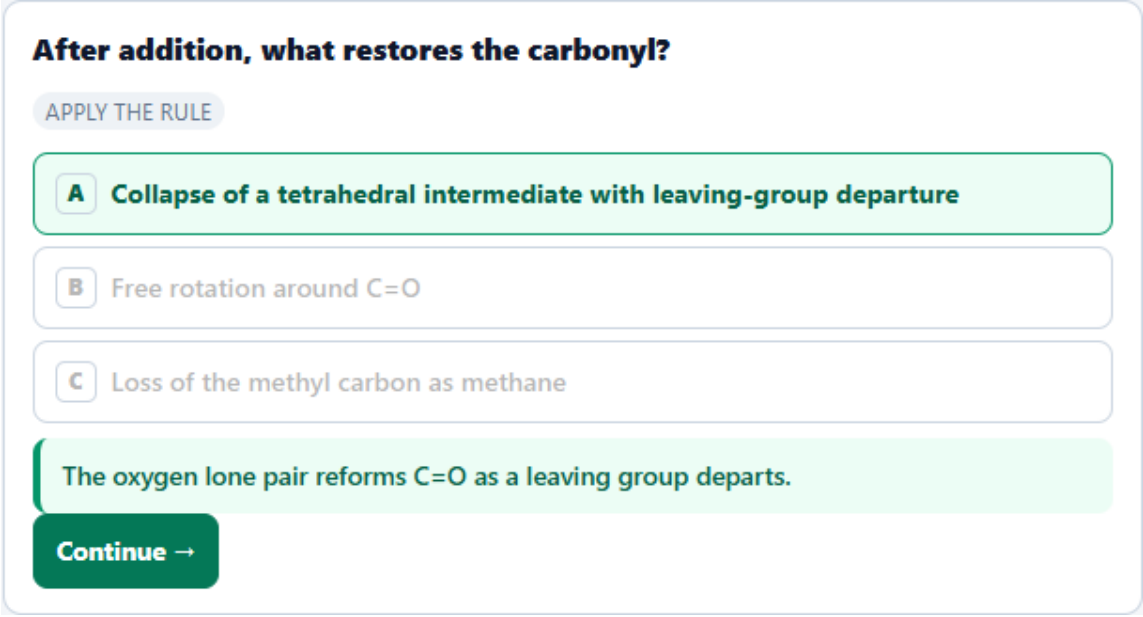
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: The acyl carbon must return to three sigma directions.

Hint 2: Reform the pi bond and expel the heteroatom leaving group.

Explanation: The oxygen lone pair reforms C=O as a leaving group departs.

Screenshot: acyl-order-stage-2-feedback



acyl-order:stage-3 | Among comparable ordinary anhydrides, esters, and amides, which is least reactive toward acyl substitution?

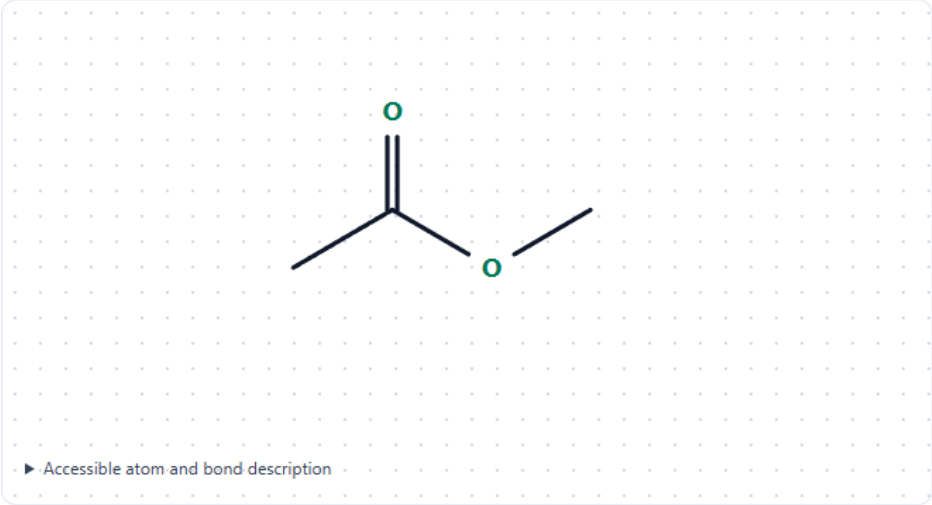
Answer type: choice

Option 0 (A): Anhydride

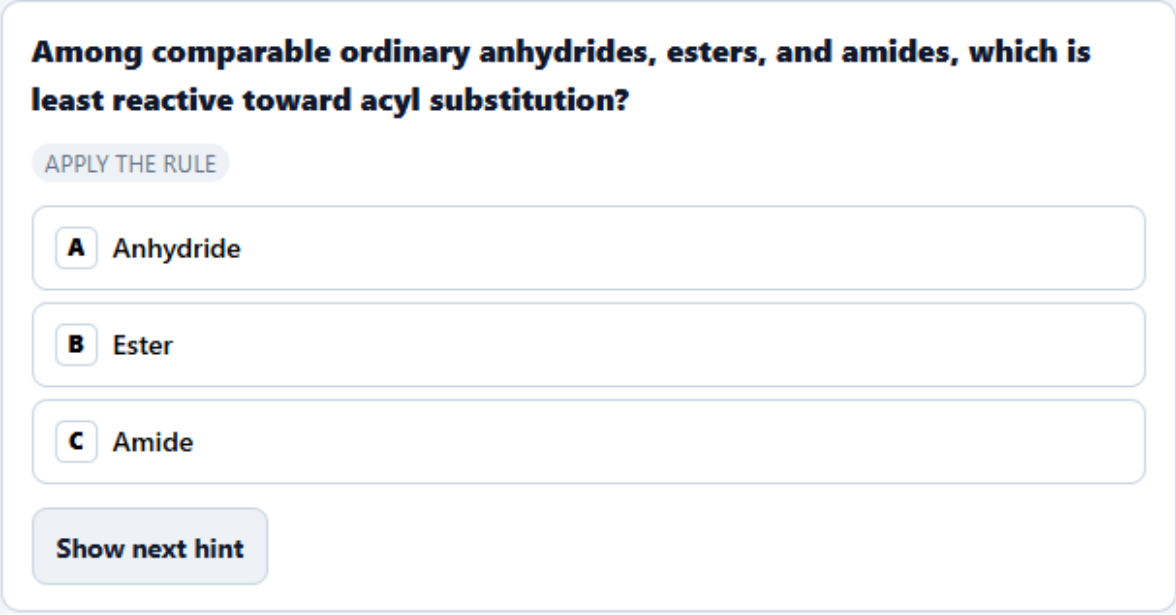
Option 1 (B): Ester

Option 2 (C): Amide

Screenshot: acyl-order-stage-3-molecule



Screenshot: acyl-order-stage-3-question



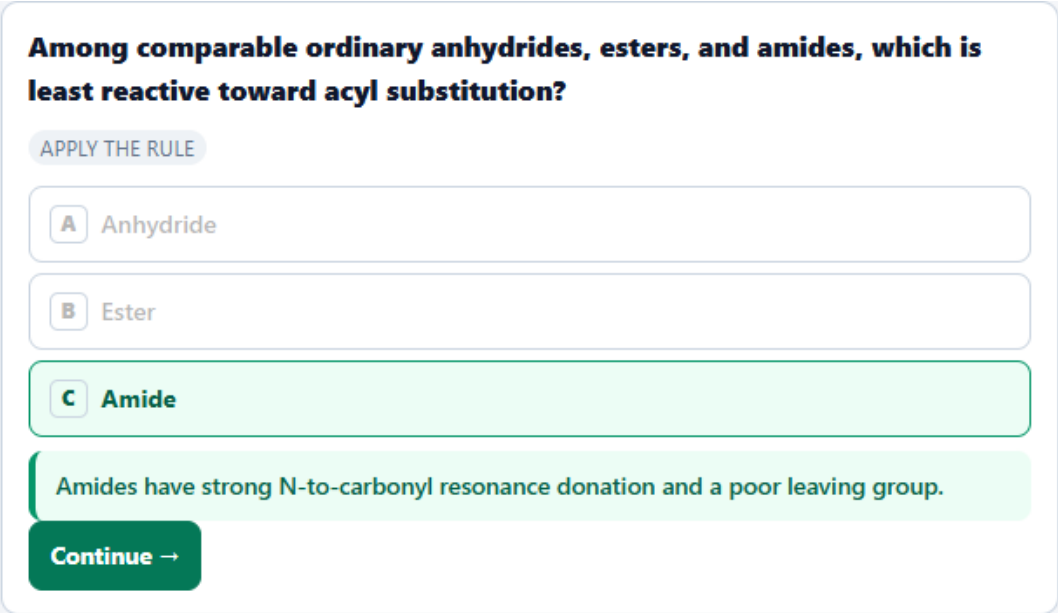
{"authorAnswer": 2, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Consider lone-pair donation and leaving-group basicity.

Hint 2: Nitrogen strongly stabilizes the amide system.

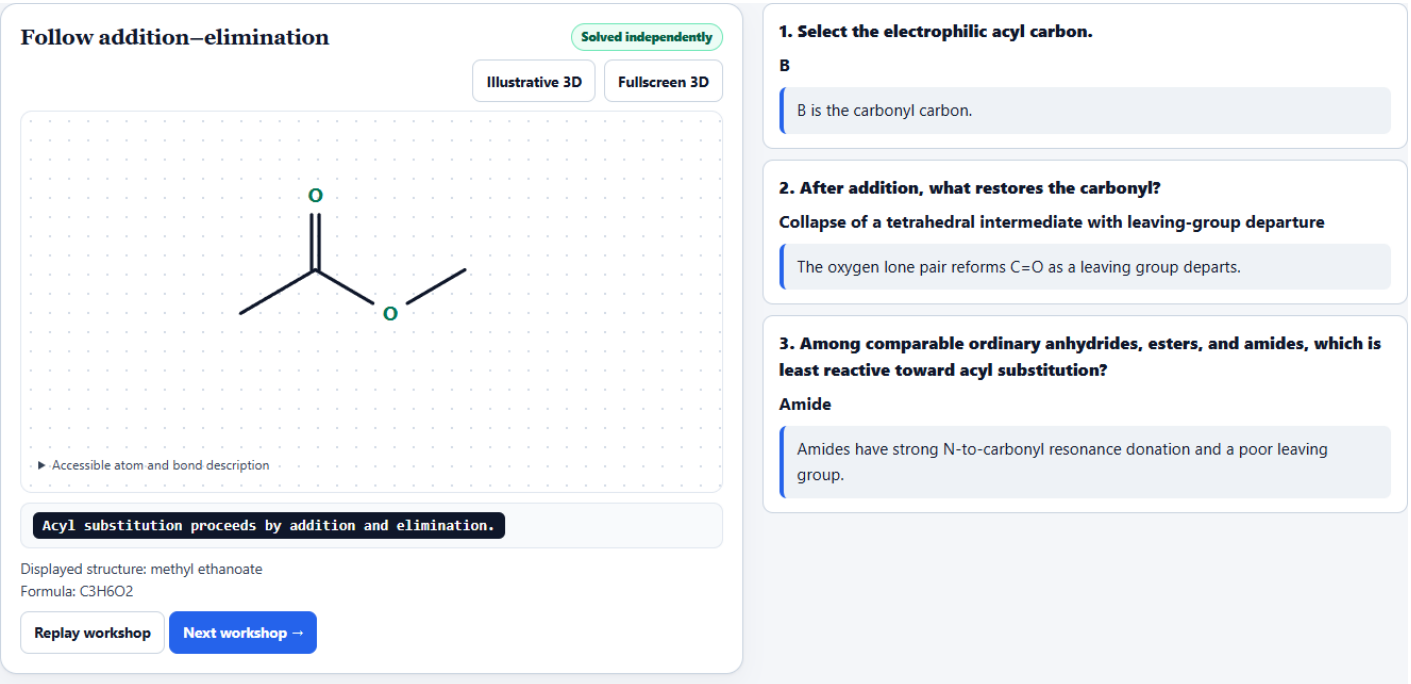
Explanation: Amides have strong N-to-carbonyl resonance donation and a poor leaving group.

Screenshot: acyl-order-stage-3-feedback



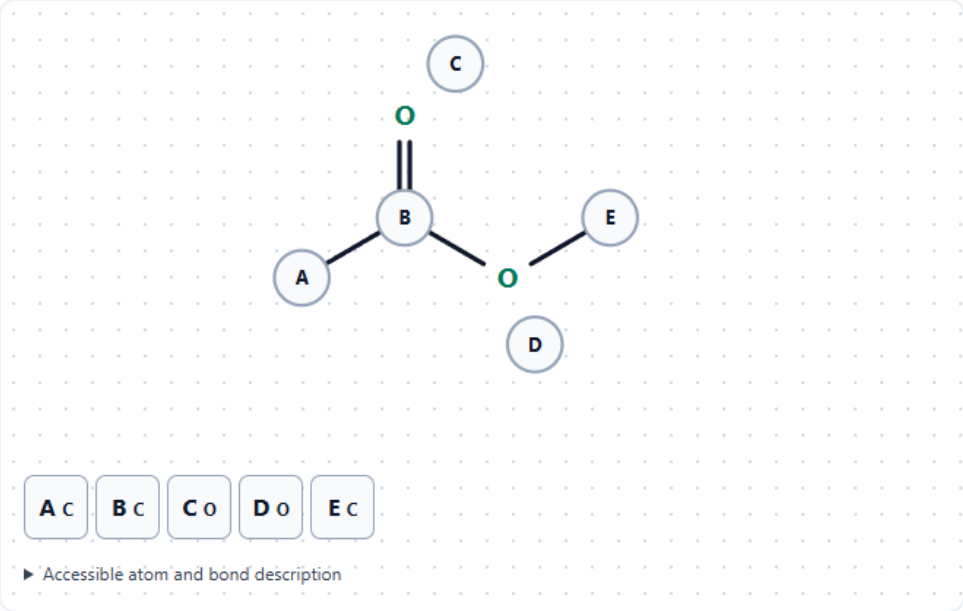
Completion summary: Acyl substitution proceeds by addition and elimination.

Screenshot: acyl-order-complete



hydrolysis-products | Track hydrolysis and workup

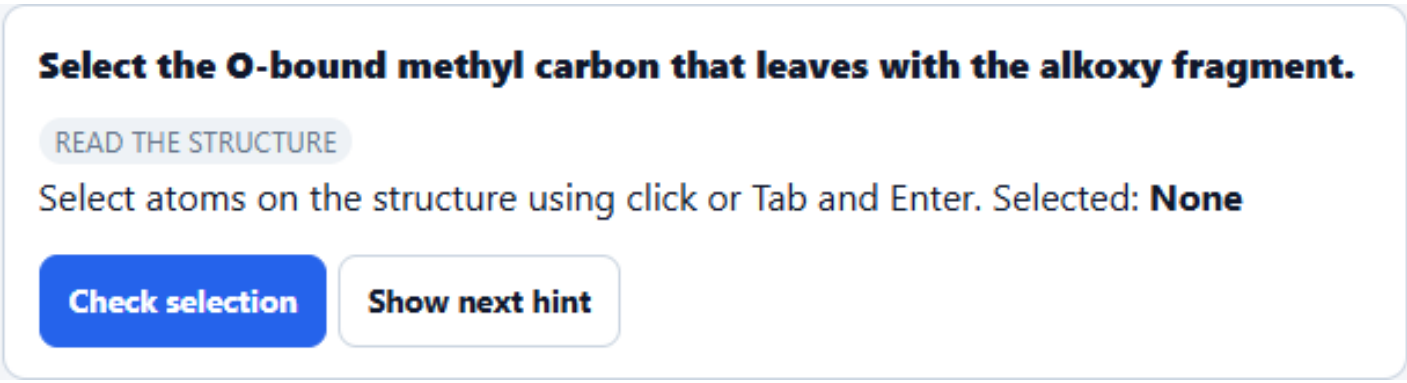
Screenshot: hydrolysis-products-stage-1-molecule



hydrolysis-products:stage-1 | Select the O-bound methyl carbon that leaves with the alkoxy fragment.

Answer type: atoms

Screenshot: hydrolysis-products-stage-1-question



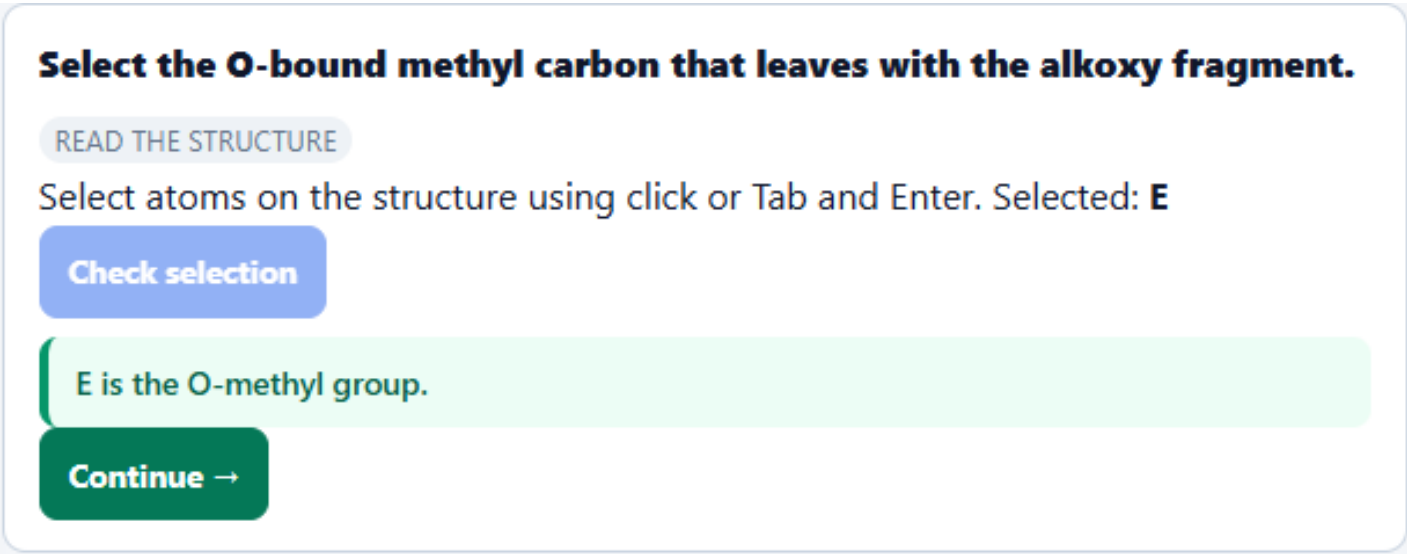
{"authorAnswer": [4], "tolerance": null, "highlightsAfterCorrectAnswer": {"sub": [3, 4]}}

Hint 1: Distinguish the acyl methyl from the O-methyl.

Hint 2: Follow the single-bonded oxygen away from the carbonyl.

Explanation: E is the O-methyl group.

Screenshot: hydrolysis-products-stage-1-feedback



hydrolysis-products:stage-2 | Under excess aqueous NaOH, the major acid-derived product is…

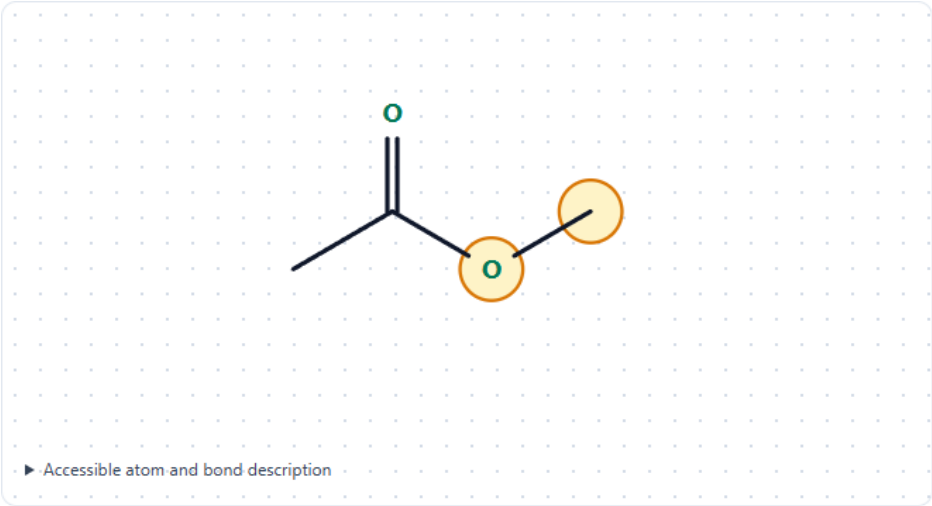
Answer type: choice

Option 0 (A): Acetate (ethanoate)

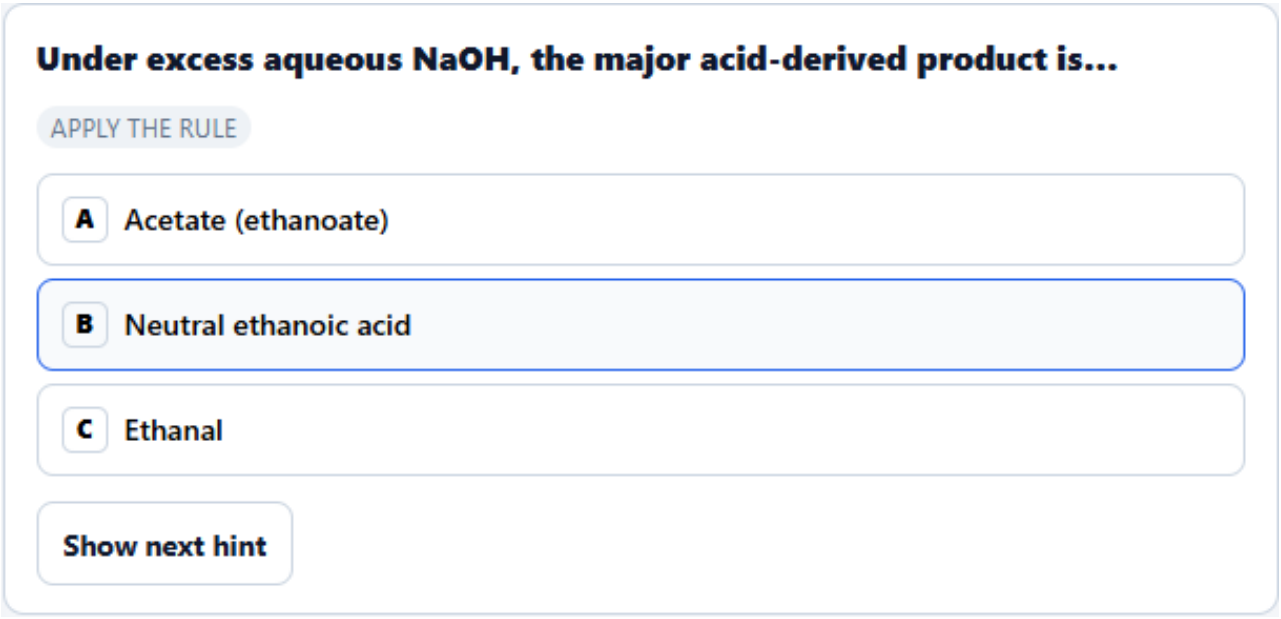
Option 1 (B): Neutral ethanoic acid

Option 2 (C): Ethanal

Screenshot: hydrolysis-products-stage-2-molecule



Screenshot: hydrolysis-products-stage-2-question



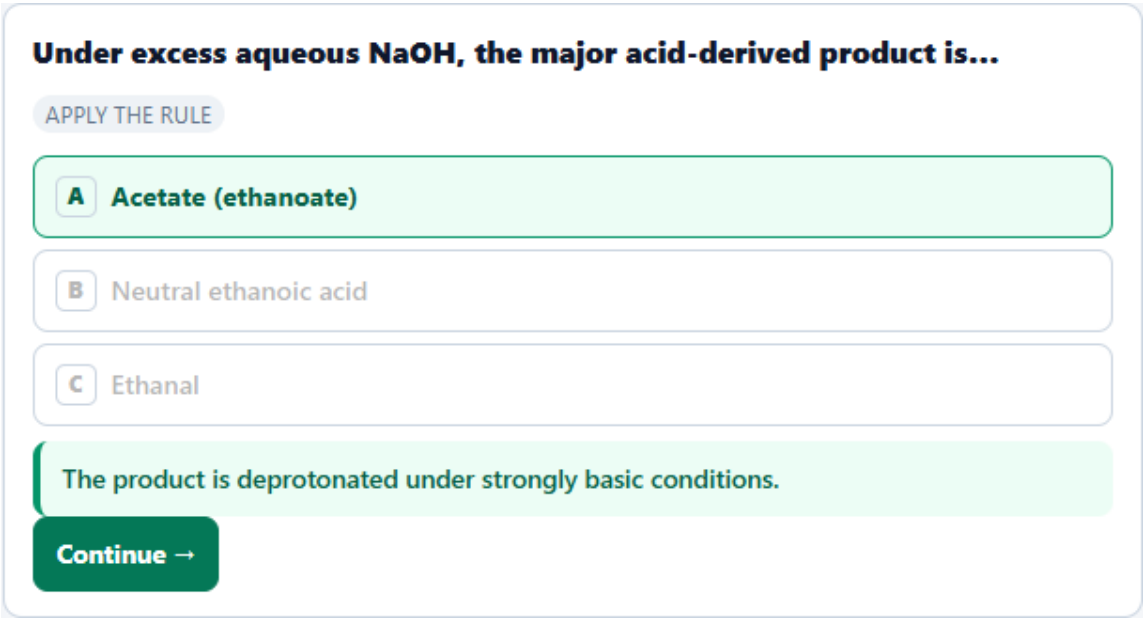
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Account for acid–base chemistry after substitution.

Hint 2: Carboxylic acids have much lower pKa than water.

Explanation: The product is deprotonated under strongly basic conditions.

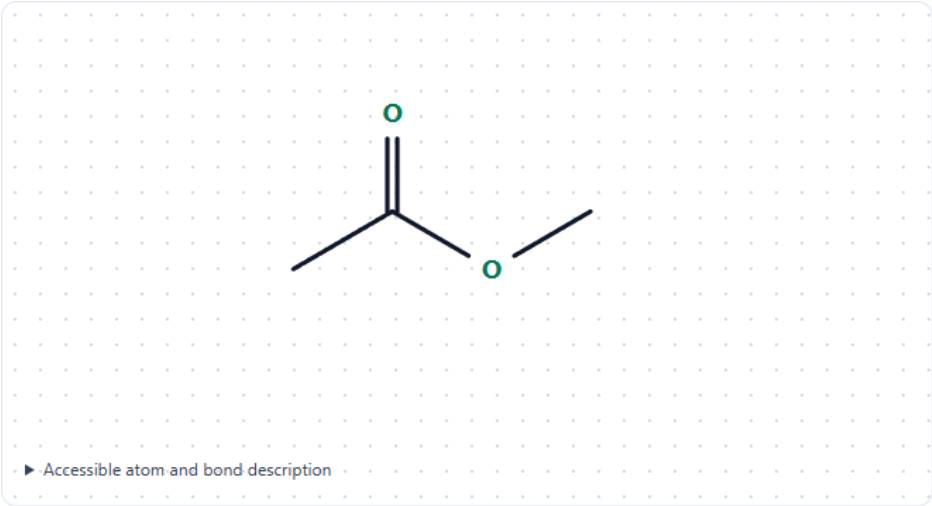
Screenshot: hydrolysis-products-stage-2-feedback



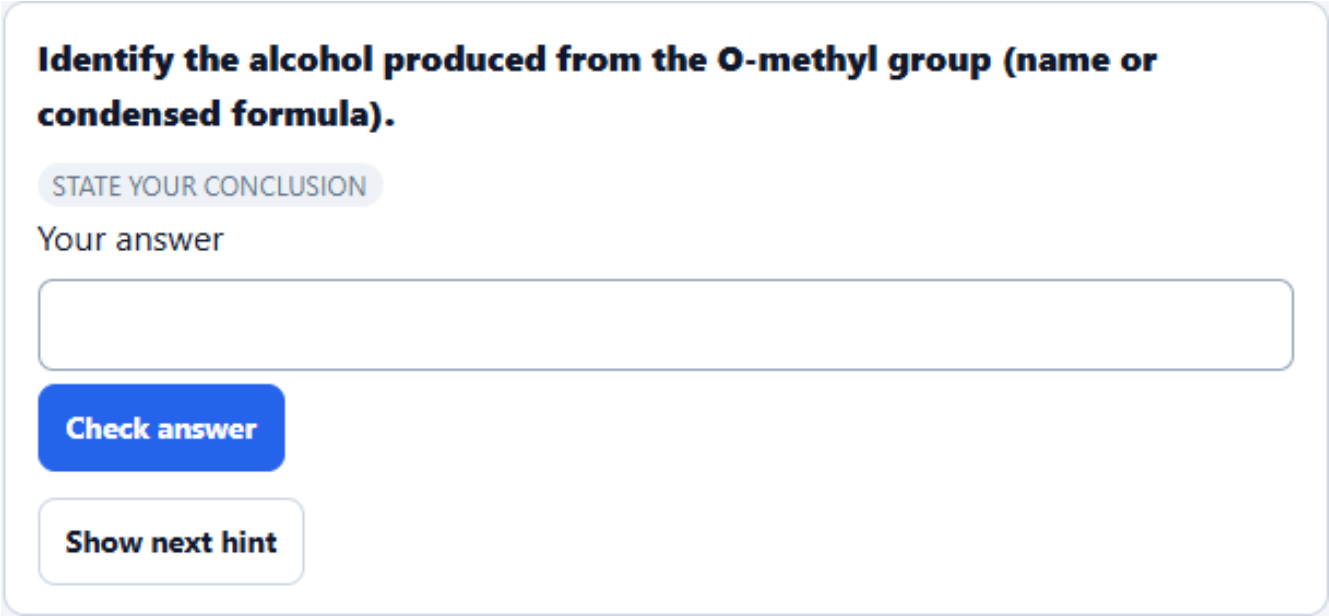
hydrolysis-products:stage-3 | Identify the alcohol produced from the O-methyl group (name or condensed formula).

Answer type: text

Screenshot: hydrolysis-products-stage-3-molecule



Screenshot: hydrolysis-products-stage-3-question



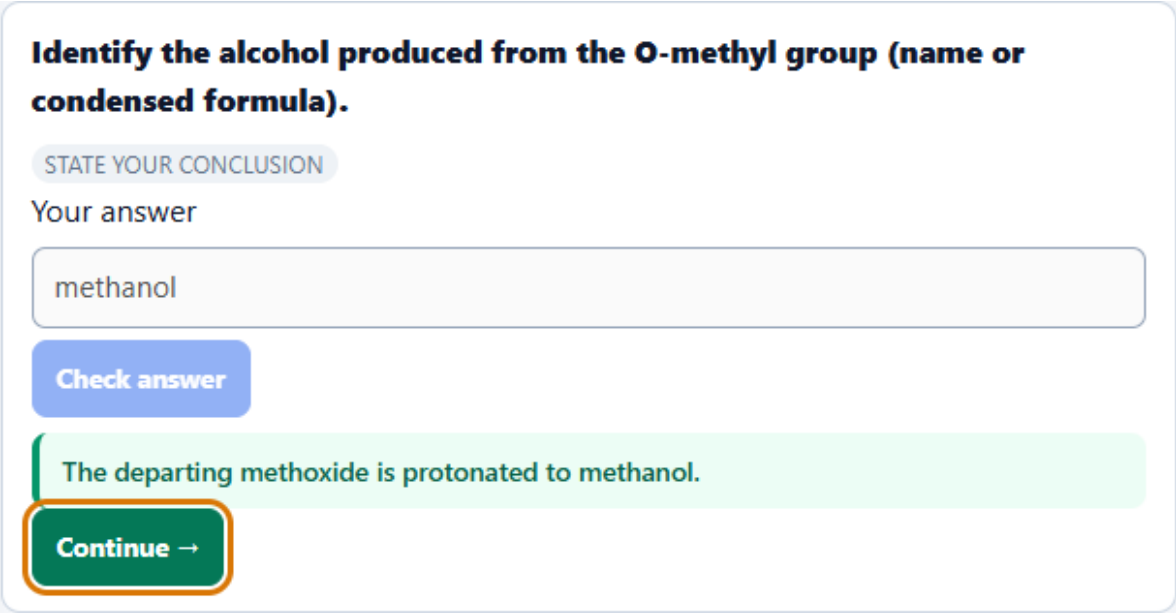
{"authorAnswer": ["methanol", "methyl alcohol", "CH3OH", "CH3-OH", "H3COH", "H3C-OH", "HOCH3", "HO-CH3", "MeOH", "Me-OH", "carbinol"], "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Keep the one-carbon alkoxy fragment.

Hint 2: Replace its bond to acyl carbon with H on oxygen.

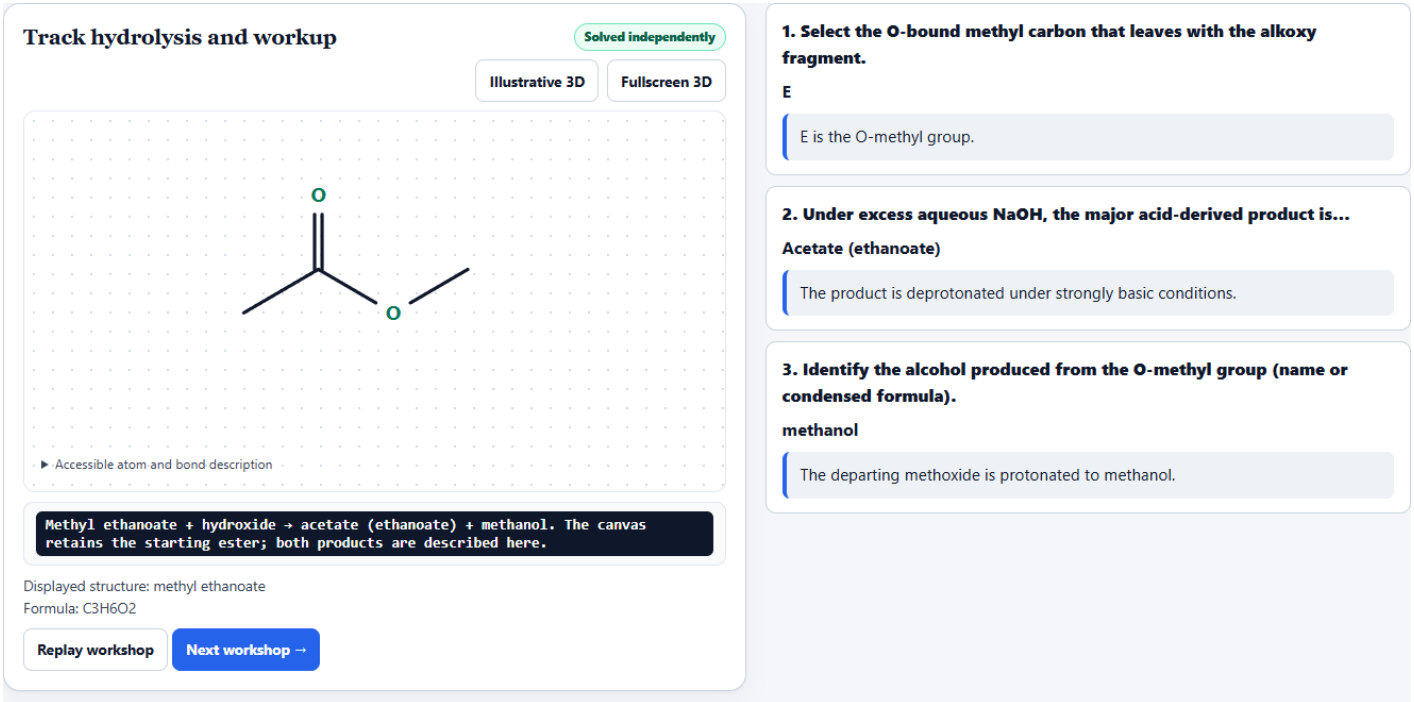
Explanation: The departing methoxide is protonated to methanol.

Screenshot: hydrolysis-products-stage-3-feedback



Completion summary: Methyl ethanoate + hydroxide → acetate (ethanoate) + methanol. The canvas retains the starting ester; both products are described here.

Screenshot: hydrolysis-products-complete



Molecular repository: author claims to verify

ester

Author claims: ethyl propanoate | formula C5H10O2 | canonical composition C5H10O2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=48.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=102.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=156.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=156.00, y=52.00; charge=0; explicit H override=none

4 (E): O; x=208.00, y=142.00; charge=0; explicit H override=none

5 (F): C; x=258.00, y=112.00; charge=0; explicit H override=none

6 (G): C; x=308.00, y=142.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 2], [2, 4, 1], [4, 5, 1], [5, 6, 1]]

betalactam

Author claims: azetidin-2-one (beta-lactam) | formula C3H5NO | canonical composition C3H5NO | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): N; x=100.00, y=60.00; charge=0; explicit H override=none

1 (B): C; x=185.00, y=60.00; charge=0; explicit H override=none

2 (C): C; x=185.00, y=145.00; charge=0; explicit H override=none

3 (D): C; x=100.00, y=145.00; charge=0; explicit H override=none

4 (E): O; x=240.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 0, 1], [1, 4, 2]]

lactone

Author claims: oxolan-2-one (gamma-butyrolactone) | formula C4H6O2 | canonical composition C4H6O2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): O; x=135.00, y=35.00; charge=0; explicit H override=none

1 (B): C; x=205.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=180.00, y=165.00; charge=0; explicit H override=none

3 (D): C; x=90.00, y=165.00; charge=0; explicit H override=none

4 (E): C; x=65.00, y=85.00; charge=0; explicit H override=none

5 (F): O; x=265.00, y=55.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 4, 1], [4, 0, 1], [1, 5, 2]]

methylacetate

Author claims: methyl ethanoate | formula C3H6O2 | canonical composition C3H6O2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=55.00, y=120.00; charge=0; explicit H override=none

1 (B): C; x=115.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=115.00, y=25.00; charge=0; explicit H override=none

3 (D): O; x=175.00, y=120.00; charge=0; explicit H override=none

4 (E): C; x=235.00, y=85.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [1, 3, 1], [3, 4, 1]]

ester-intermediate

Author claims: tetrahedral intermediate of methyl ethanoate hydroxide addition | formula C3H7O3 | canonical composition C3H7O3 | net charge -1

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=55.00, y=145.00; charge=0; explicit H override=none

1 (B): C; x=115.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=175.00, y=25.00; charge=-1; explicit H override=none

3 (D): O; x=175.00, y=145.00; charge=0; explicit H override=none

4 (E): C; x=235.00, y=110.00; charge=0; explicit H override=none

5 (F): O; x=55.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 1], [3, 4, 1], [1, 5, 1]]

anhyd

Author claims: ethanoic anhydride | formula C4H6O3 | canonical composition C4H6O3 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=45.00, y=130.00; charge=0; explicit H override=none

1 (B): C; x=100.00, y=96.00; charge=0; explicit H override=none

2 (C): O; x=155.00, y=130.00; charge=0; explicit H override=none

3 (D): C; x=210.00, y=96.00; charge=0; explicit H override=none

4 (E): C; x=265.00, y=130.00; charge=0; explicit H override=none

5 (F): O; x=100.00, y=40.00; charge=0; explicit H override=none

6 (G): O; x=210.00, y=40.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 4, 1], [1, 5, 2], [3, 6, 2]]

acetamide

Author claims: ethanamide | formula C2H5NO | canonical composition C2H5NO | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=55.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=115.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=115.00, y=25.00; charge=0; explicit H override=none

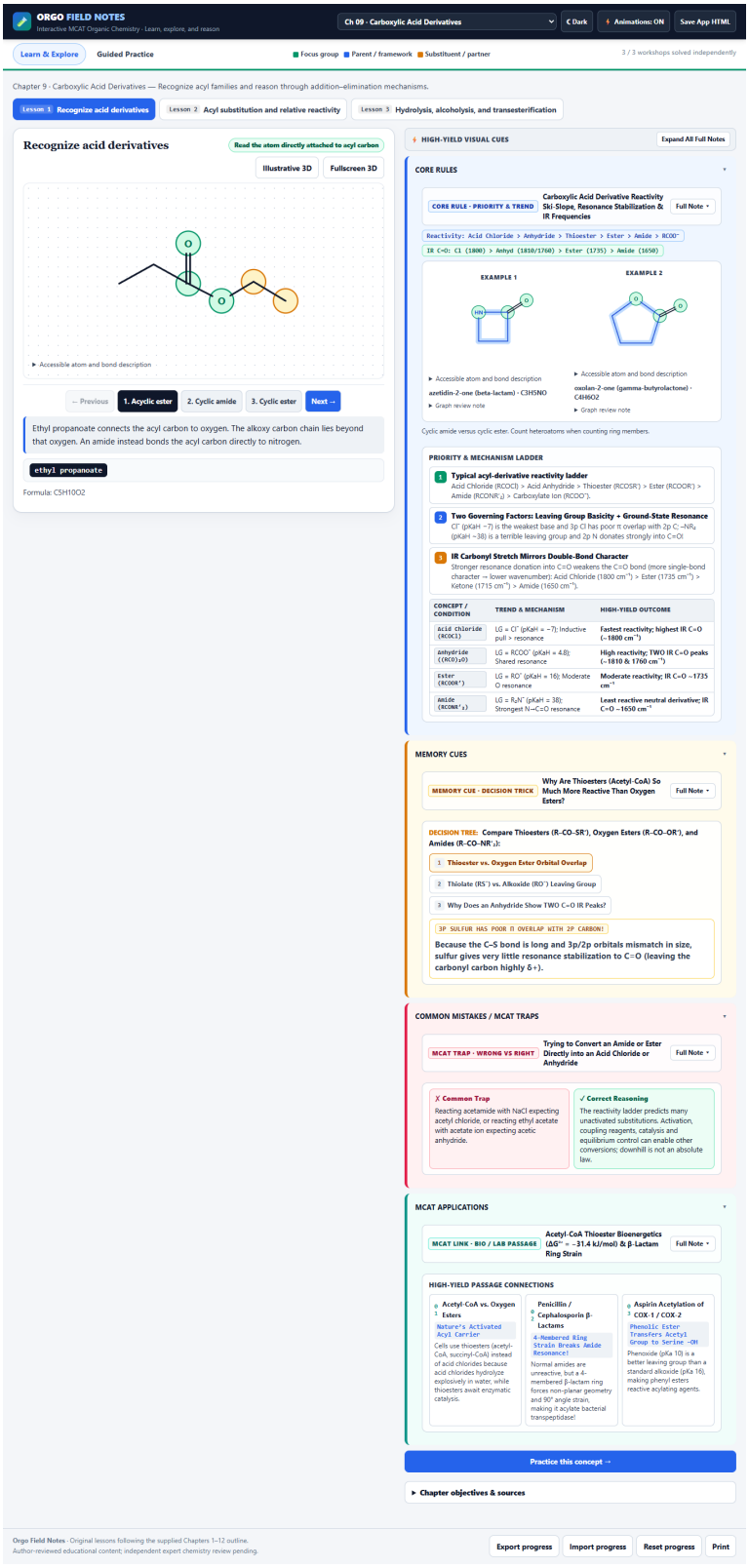
3 (D): N; x=175.00, y=115.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [1, 3, 1]]

Layout samples

These overview images are scaled for page fit. Detailed structure and stage images appear earlier. Samples cover the first concept on desktop and mobile in both themes, not every responsive state.

Screenshot: layout-desktop-light



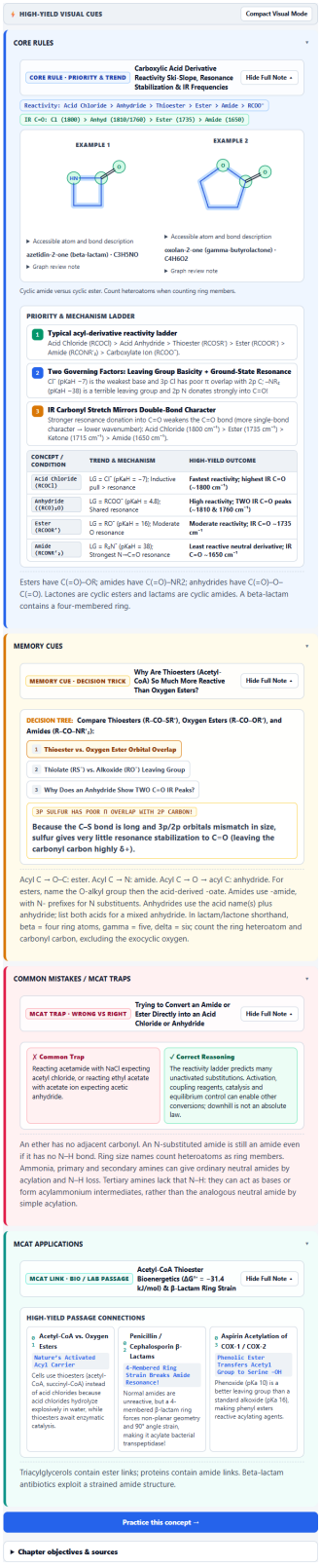
Screenshot: layout-mobile-light



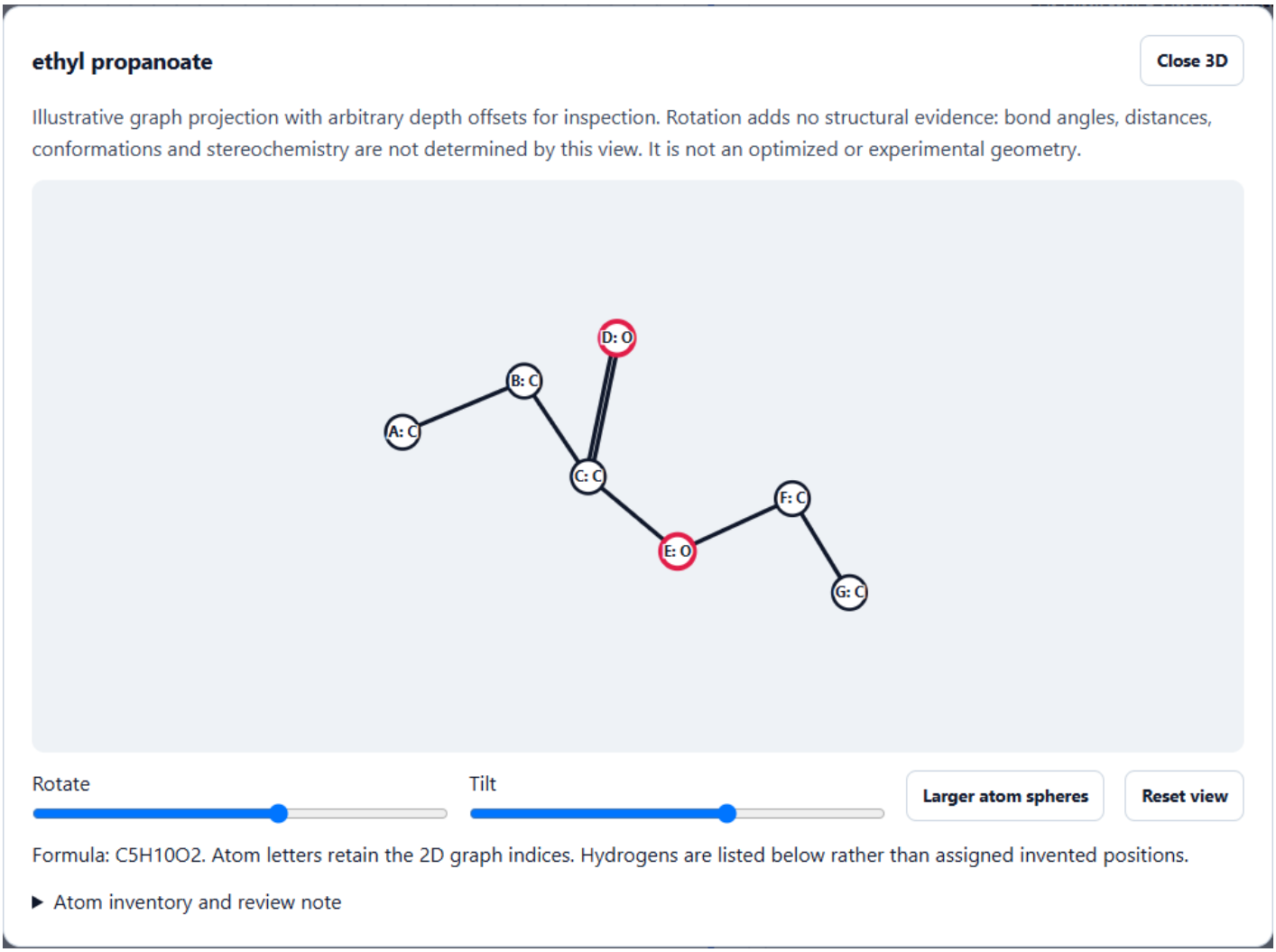
Screenshot: layout-mobile-dark



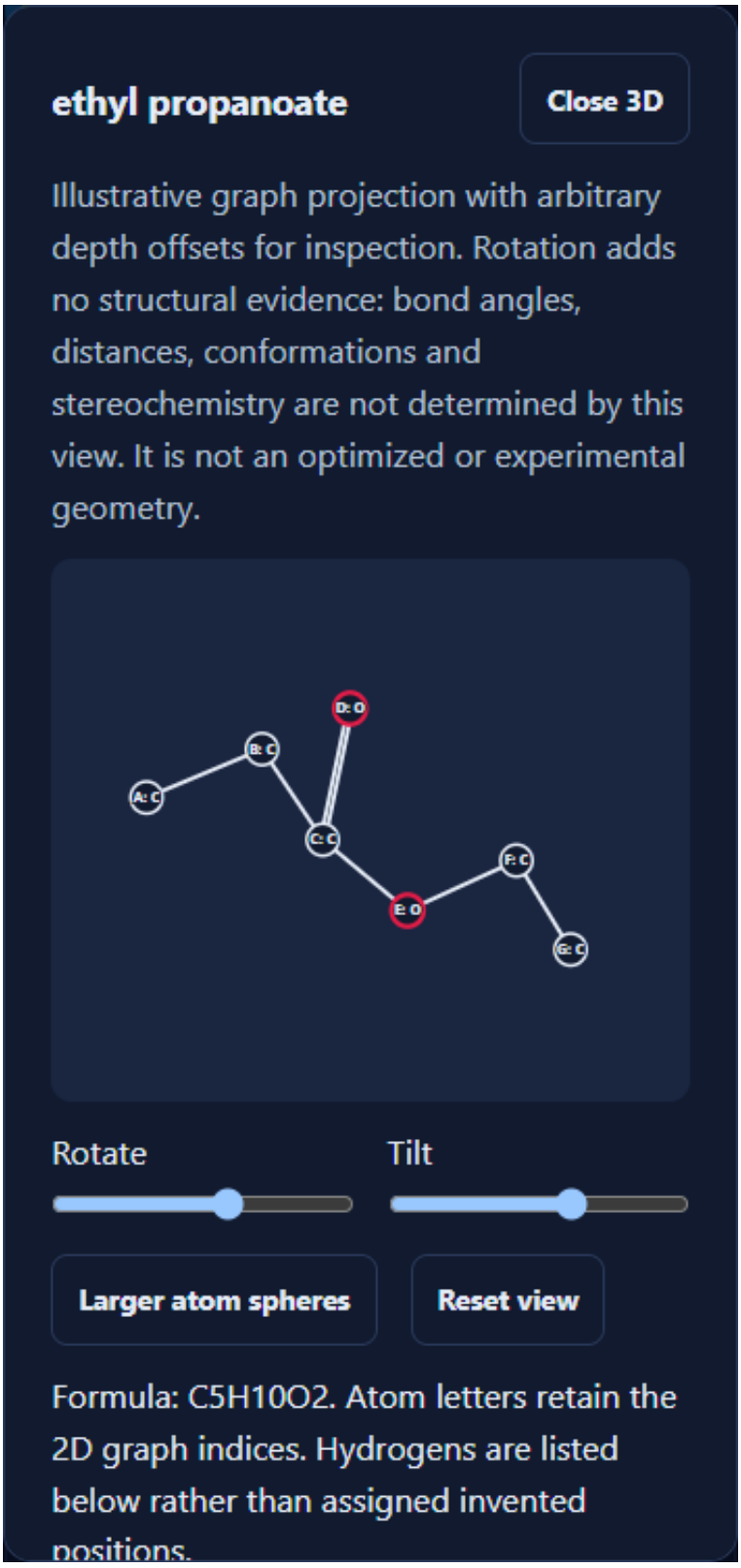
Screenshot: visual-notes-expanded



Screenshot: projection-desktop-light



Screenshot: projection-mobile-dark



Execution evidence

48 screenshots and 9 recorded interactions. Incorrect answers rejected and encoded keys accepted for captured guided stages. This checks implementation against its data; it does not validate chemistry independently. Static screenshots do not verify animation timing, arbitrary clicks, or screen-reader experience.